BIOLOGY GRADE 10: CHEMICALS OF LIFE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.2 (6 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Biology
Strand	Strand 1.0: Cell Biology and Biodiversity
Sub-Strand	Sub-Strand 1.2: Chemicals of Life
Total Duration	6 lessons × 40 minutes = 240 minutes total
Sub-Strand Content	– Carbohydrates — structure, types (monosaccharides, disaccharides, polysaccharides) and functions in living organisms – Proteins — structure (amino acids, peptide bonds), types and biological functions – Lipids — structure (fatty acids, glycerol), types (fats, oils, waxes, phospholipids) and functions – Nucleic acids — DNA and RNA structure and their roles in heredity and protein synthesis – Vitamins and minerals — classification, sources (local Kenyan foods), deficiency diseases and functions – Water — properties and biological importance in living cells – Food tests — chemical tests for reducing sugars (Benedict's), starch (iodine), proteins (Biuret), lipids (Sudan III / ethanol emulsion) using locally available food samples (ugali, beans, milk, avocado, sukuma wiki)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) identify the major chemicals of life (carbohydrates, proteins, lipids, nucleic acids, vitamins, minerals and water) and describe their structures, b) explain the functions of the major chemicals of life in living organisms, c) carry out food tests to identify carbohydrates, proteins and lipids in food substances, d) relate deficiency of specific chemicals of life to diseases and health conditions observed in the community, e) appreciate the importance of a balanced diet containing the chemicals of life for good health and wellbeing.
Core Competencies	– Communication and Collaboration: Learners work in small groups to carry out food tests on local Kenyan food samples (ugali, beans, avocado, milk), record results in shared tables and present findings to the class, developing scientific communication skills. – Critical Thinking and Problem Solving: Learners analyse food test results, identify unknown substances in food samples and propose explanations for deficiency diseases observed in Kenyan communities, developing evidence-based reasoning. – Imagination and Creativity: Learners design and construct visual molecular models of carbohydrates, proteins and lipids using locally available materials (bottle tops, straws, clay) to represent biological macromolecules. – Self-efficacy: Learners develop confidence in handling laboratory apparatus and interpreting chemical test results independently, building a sense of scientific competence.
Core Values	– Responsibility: The learner safely handles chemicals and laboratory apparatus during food tests and disposes of materials responsibly to protect the school environment. – Respect: The learner appreciates diverse dietary practices and food choices across Kenyan communities when discussing

	sources of the chemicals of life. – Unity: The learner collaborates equitably with peers of different backgrounds during group practical activities and model-building tasks. – Integrity: The learner records food test observations honestly and accurately, even when results do not match predictions.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners generate questions about why different foods cause different health outcomes in Kenyan communities, anchoring inquiry into the chemical composition of food. – Planning and Carrying Out Investigations: Learners plan and conduct systematic food tests (Benedict's, iodine, Biuret, ethanol emulsion) on a range of local food samples, controlling variables and recording observations. – Analysing and Interpreting Data: Learners analyse colour changes from food tests, construct results tables and draw conclusions about the chemical composition of foods such as ugali (starch), nyama choma (protein) and avocado (lipids). – Constructing Explanations: Learners use evidence from food tests and molecular structure diagrams to explain how the chemical composition of food relates to its biological function and to observed deficiency diseases. – Developing and Using Models: Learners build and revise physical models of biological macromolecules (monosaccharide chains, amino acid sequences) using craft materials to visualise structure–function relationships.
Pertinent & Contemporary Issues (PCIs)	– Health Education: The learner relates the chemicals of life to nutrition, dietary balance and prevention of deficiency diseases common in Kenya (e.g., kwashiorkor, marasmus, scurvy, rickets, anaemia), promoting informed health choices. – Environmental Conservation: The learner uses locally available and recycled materials (bottle tops, straws, clay) to build molecular models, minimising waste and promoting sustainable resource use. – Safety and Security: The learner observes safe laboratory practices when handling chemicals (Benedict's reagent, iodine solution, Biuret reagent) during food tests, protecting personal and peer safety. – Financial Literacy: The learner evaluates the nutritional value of affordable, locally available Kenyan foods to make cost-effective dietary decisions that meet the body's chemical requirements.
Career Connections	– Nutritionist / Dietitian: Understanding the chemicals of life and their functions in the body is foundational to advising individuals and communities on healthy, balanced diets using locally available Kenyan foods. – Biochemist: The study of carbohydrates, proteins, lipids and nucleic acids at the molecular level is the core of biochemistry research in institutions such as the Kenya Medical Research Institute (KEMRI). – Medical Laboratory Technologist: Performing chemical food tests mirrors the diagnostic laboratory tests used in hospitals across Kenya to analyse blood composition, urine and tissue samples. – Food Scientist / Technologist: Knowledge of food chemistry is essential for food processing, quality control and product development in Kenya's agri-food industry (e.g., maize milling, dairy processing in Eldoret). – Pharmacist: Understanding proteins, lipids and nucleic acids is critical for drug design, including vaccines and biologics produced or distributed in Kenya. – Agricultural Scientist: Relating soil minerals and plant biochemistry to crop nutritional value supports agricultural research at institutions such as KARI (Kenya Agricultural and Livestock Research Organization).
Focus for Lessons	This unit investigates the major biological macromolecules — carbohydrates, proteins, lipids and nucleic acids — alongside vitamins, minerals and water, and their essential roles in living organisms. Using a phenomenon-driven, hands-on approach, learners conduct chemical food tests on familiar Kenyan foods (ugali, beans, avocado, nyama choma, sukuma wiki) to discover what these foods are made of at the molecular level and why that matters for health. The unit connects molecular biology to real-world nutritional challenges in Kenyan communities, empowering learners to make evidence-based decisions about diet and health.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do two children in Kisumu eating the same amount of food every day end up with completely different

	bodies — and what does that tell us about what food is really made of? KICD KEY INQUIRY QUESTIONS: 1. What are the chemicals of life and what roles do they play in living organisms? 2. How can we identify the chemicals present in different foods using chemical tests?
Anchoring Phenomenon	A Standard 5 pupil from Kisumu develops a swollen belly, thin limbs, discoloured hair and extreme fatigue despite eating ugali (maize starch) every day. Meanwhile, a child in the same village who eats githeri (maize and beans) regularly appears healthy and energetic. Both families have access to the same amount of food in terms of quantity, yet their health outcomes are strikingly different. Students are shown photographs of the two children and asked: how can two children eating similar amounts of food every day have such different bodies and health? This puzzling observation — that not all food is the same at the chemical level — anchors the entire unit. It drives learners to investigate what food is actually made of (the chemicals of life), what each chemical does in the body, and why the absence of even one type of chemical can cause visible, devastating changes to a living organism. The phenomenon is deliberately chosen because kwashiorkor (protein deficiency) is a recognisable condition in Kenyan communities, making the science personally meaningful and socially relevant.
Storyline Thread	Lesson 1 — Predict: What Is Food Really Made Of? Students observe photographs of the two Kisumu children and the anchoring phenomenon is introduced. In pairs, learners predict what they think makes up food at the molecular (chemical) level, and why the same quantity of different foods causes different health outcomes. A class Driving Question Board (DQB) is initiated. The teacher introduces the six major chemicals of life (carbohydrates, proteins, lipids, nucleic acids, vitamins/minerals, water) as the framework for the unit's investigation. Lesson 2 — Observe (Part 1): Carbohydrates and Proteins — Structure and Function Learners investigate the structure of carbohydrates (mono-, di-, polysaccharides) and proteins (amino acids, peptide bonds) through annotated diagrams, molecular model-building using bottle tops and straws, and guided reading. They connect glucose → glycogen storage and amino acids → body-building proteins to the anchoring phenomenon (why ugali alone cannot keep the Kisumu child healthy). Food sources from the Kenyan diet are mapped to each chemical class. Lesson 3 — Observe (Part 2): Lipids, Nucleic Acids, Vitamins, Minerals and Water Learners extend their investigation to lipids (fatty acids + glycerol), DNA/RNA (brief structural overview), vitamins, minerals and water. They use a comparison table to record the structure, food sources (avocado, milk, sukuma wiki, fish) and functions of each chemical. The supporting phenomenon of cooking oil and water not mixing is used to explore the hydrophobic nature of lipids. The role of water as a universal solvent in cells is emphasised. Lesson 4 — Observe (Part 3): Food Tests Practical — Identifying Chemicals in Kenyan Foods Learners plan and carry out food tests on a set of local food samples (ugali/maize flour for starch, beans extract for protein, avocado for lipids, orange juice for vitamin C using DCPIP, glucose solution for reducing sugars). Tests used: iodine (starch), Benedict's reagent (reducing sugars), Biuret reagent (protein), ethanol emulsion/Sudan III (lipids). Learners record colour changes, construct a results table and draw conclusions about the chemical composition of each food. Lesson 5 — Explain: Sensemaking — Connecting Chemistry to Health Learners use their food test data and molecular knowledge to construct scientific explanations for the anchoring phenomenon: the child eating only ugali gets carbohydrates but is severely protein-deficient (kwashiorkor), while the githeri-eating child receives both carbohydrates and proteins. Learners match deficiency diseases (kwashiorkor, marasmus, scurvy, rickets, anaemia) to the missing

	chemical of life. Groups present their explanations and update the DQB with new understanding. The teacher facilitates a class discussion connecting findings to Kenyan public health nutrition campaigns. Lesson 6 — Model Building and DQB Consolidation: Building Our "Chemicals of Life" Model Learners revise and finalise a cumulative visual model (poster or digital diagram) showing the six chemicals of life, their molecular structures, functions, Kenyan food sources and consequences of deficiency — all anchored back to the two children from Kisumu. Each group presents their model, explains what has changed from their Lesson 1 predictions, and identifies remaining questions. The DQB is reviewed and answered questions are celebrated. A formative assessment task asks learners to evaluate a typical Kenyan daily menu and identify which chemicals of life are present, missing or insufficient.
Supporting Phenomena	– A Kericho tea farmer notices that maize plants growing on nitrogen-poor soil are stunted and pale yellow, while maize on fertilised soil is tall and dark green — connecting mineral/protein deficiency in plants to the same chemical principles studied in human nutrition. – A student observes that cooking oil (lipid) and water do not mix when making a salad dressing at home in Nairobi, puzzling over why these two common food substances refuse to combine — introducing the hydrophobic nature of lipids. – A runner from Eldoret who eats a high-carbohydrate meal (ugali and beans) before a race performs better than one who skipped the meal, raising the question of how the body converts food chemicals into usable energy. – A preserved specimen jar in the school laboratory contains an animal tissue that has remained unchanged for weeks, prompting students to ask what chemicals are responsible for the structure and preservation of living tissue.

LESSON 1 (80 minutes (double lesson)): Predict: What Is Food Really Made Of?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon and elicit students' prior ideas about what food is made of, establishing the need to investigate the chemicals present in different foods.
Knowledge	By the end of this lesson, the learner should be able to identify that food is composed of different chemical substances (carbohydrates, proteins, lipids, vitamins, minerals, water) and that different foods contain different proportions of these chemicals.
Skills	By the end of this lesson, the learner should be able to construct an initial predictive model showing what chemicals they think are present in ugali and githeri, and articulate a testable question arising from the anchoring phenomenon.
Attitudes	By the end of this lesson, the learner should appreciate that scientific questions arise from real, observable puzzles in their own communities, and demonstrate curiosity and empathy when engaging with evidence of nutritional disease in Kenyan children.
Key Inquiry Question	What are the chemicals of life and what roles do they play in living organisms?
Purpose in Storyline	This is the opening lesson of the Chemicals of Life unit. It launches the anchoring phenomenon — two children in Kisumu with dramatically different health outcomes despite eating similar quantities of food — and drives learners to ask: what is food really made of at the chemical level? Students make initial predictions, open the Driving Question Board, and build a first draft of their explanatory model. Every subsequent lesson will return to this phenomenon to deepen and revise their understanding.
Safety Notes	No hazardous chemicals are used in this lesson. Photographs displayed must be handled with sensitivity — remind learners that the children shown represent real conditions affecting Kenyan families and should be discussed with respect and empathy, not mockery. If any learner shows signs of personal distress related to food insecurity, the teacher should respond with discretion and follow the school's pastoral care procedures.

B. LESSON OVERVIEW
Students are introduced to the anchoring phenomenon: a child in Kisumu who eats ugali daily has developed kwashiorkor (swollen belly, thin limbs, discoloured hair, extreme fatigue), while a child in the same village who eats githeri regularly is healthy and energetic. Both children consume similar quantities of food. Students examine photographs, share initial reactions, and make structured predictions about what ugali and githeri are made of at the chemical level. They record their predictions in their science notebooks as annotated diagrams and then co-construct the first entries on the class Driving Question Board (DQB). The lesson closes with students drafting their first explanatory model — a simple before-and-after diagram linking food chemicals to body outcomes. No chemical tests are performed in this lesson; the purpose is entirely to surface prior knowledge, generate curiosity, and anchor the unit question.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Dehydration](#)	Students enter the classroom to find two large photographs projected or printed and displayed	As students enter, say nothing — allow the photographs to speak first. After 60 seconds of silent	Strategy: Notice & Wonder + Annotated Predictive Diagram. This strategy surfaces students'	Observable indicator: Every student has a written prediction ('I think… because…') and a labelled

synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Eukaryotes to Multicellular Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	at the front: Photo A shows a child from Kisumu with visible signs of kwashiorkor (swollen abdomen, thin limbs, reddish discoloured hair, lethargic posture); Photo B shows a healthy, energetic child of the same age from the same village. A caption beneath both reads: 'Both children eat approximately the same amount of food every day. Both live in the same village near Kisumu.' Students observe silently for 60 seconds, then write a 'Notice & Wonder' entry in their science notebooks: three things they NOTICE and three things they WONDER. In pairs, students share their notices and wonders. The teacher then reveals the food details: Child A eats ugali (maize starch) as the main meal daily; Child B eats githeri (maize and beans) regularly. Students individually write a prediction in response to the prompt: 'I think the difference in the children's bodies is caused by _______ because _______.' Students then sketch an annotated diagram predicting what chemicals or substances they think are inside ugali versus githeri, labelling whatever they already know or guess (e.g. starch, protein, energy, vitamins). Pairs share their diagrams and predictions with another pair (think-pair-share-square).	observation, say: 'In your notebook, write three things you NOTICE about these two children and three things you WONDER. You have 90 seconds. Go.' [WAIT 90 seconds — circulate silently, do not prompt.] After writing: 'Turn to your partner. Share one NOTICE and one WONDER each. You have 60 seconds.' [WAIT 60 seconds.] Cold-call 4 students for NOTICES: 'Amina — tell us one thing you noticed.' [WAIT 10 seconds.] Repeat for Otieno, Wanjiru, Kipchoge. Cold-call 3 students for WONDERS: 'Fatuma — what is one thing you wondered?' [WAIT 10 seconds.] Repeat for two more. Now reveal the food information. Say: 'Here is one more piece of information. Child A eats ugali — maize starch — as the main meal every day. Child B eats githeri — maize and beans — regularly. Same village. Same quantity of food. Write this in your notebook and now make a prediction: I think the difference in the children's bodies is caused by _______ because _______. You have 2 minutes.' After predictions: 'Now sketch what you think is INSIDE ugali chemically, and what is inside githeri chemically. Draw a circle for each food, and write or draw what you think is in there. Label everything — even guesses are	prior knowledge and misconceptions before instruction begins, making thinking visible. By drawing what they think is 'inside' ugali and githeri, students externalise mental models that the teacher can assess and that students themselves can revise across the unit. Connecting the prediction to a visible, emotionally resonant phenomenon (a sick child from their own region) activates both cognitive engagement and personal relevance, increasing motivation to resolve the puzzle.	predictive diagram in their notebook within 5 minutes. Teacher circulates and looks for: (1) whether students name any specific chemicals (starch, protein, vitamins) or stay at a surface level ('ugali gives energy, githeri has more things'); (2) whether students connect the chemical difference to the physical difference in the children's bodies. Teacher mentally notes 2–3 students with rich prior knowledge (to use as discussion anchors later) and 2–3 students with blank or vague diagrams (to pair strategically). No marks are given — teacher says: 'These predictions are your starting point, not your final answer. We will come back to them at the end of the unit and see how your thinking has grown.'

		valuable right now.' [WAIT 3 minutes.] Pair to square: 'Share your diagram with your partner, then both pairs share with each other. Identify one prediction you AGREE on and one you DISAGREE on.' [WAIT 3 minutes.] Close the phase: 'You have just done what every scientist does — you looked at a puzzle from your own community and asked a question. That is exactly where science begins.'		
Observe Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Eukaryotes to Multicellular Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students receive a one-page Phenomenon Card containing: (a) a short narrative about the two Kisumu children with key details about their diets and symptoms, (b) a labelled photograph of a child with kwashiorkor with symptoms annotated (swollen belly = oedema, thin limbs = muscle wasting, reddish hair = hair depigmentation, fatigue = metabolic impact), (c) a simple food comparison table showing the approximate composition of ugali vs githeri in plain language (e.g. 'Ugali is made mainly from maize flour; githeri is made from maize AND beans'), and (d) one guiding question: 'What do you think beans contain that maize alone does not?' Students read the Phenomenon Card individually (3 minutes), then annotate it — circling words they don't understand, underlining things that surprise them, and starring things that connect to their predictions. Small groups of 3–4 then discuss: 'What does this evidence tell us? What does it NOT tell us yet?'	Distribute the Phenomenon Cards face-down. Say: 'Do not turn over until I say go. When I say go, read silently and annotate: circle words you don't know, underline surprises, star things that connect to your prediction. You have 3 minutes. Go.' [WAIT 3 minutes — circulate, do not explain vocabulary yet.] After reading: 'Now in your groups of 3 or 4, discuss this question: What does this evidence SHOW us, and what does it NOT tell us yet? Record your answers in a two-column table in your notebook. You have 4 minutes.' [WAIT 4 minutes.] Circulate and listen to group talk. Select one group that has identified 'we don't know what chemicals are in beans' and one group that has connected the oedema symptom to something missing in the diet. Plan to use these groups in the debrief.	Strategy: Phenomenon Card Annotation + Two-Column Evidence Table. Annotating the Phenomenon Card trains students to read scientific descriptions critically — distinguishing what is observed from what is inferred. The two-column table ('What the evidence shows' vs 'What we still need to find out') is a core scientific practice (NGSS SEP 6: Constructing Explanations; SEP 1: Asking Questions) that makes the boundary between current knowledge and productive ignorance explicit. This explicit boundary-marking is what makes the Driving Question Board feel purposeful rather than arbitrary.	Observable indicator: Each group's two-column table has at least 2 entries in each column within 4 minutes. Teacher specifically checks whether groups are able to distinguish observation ('Child A has a swollen belly') from inference ('Child A is missing something in the food'). If groups only list observations, teacher prompts: 'What do you think is CAUSING those observations?' — not as instruction, but as a thinking prompt. Teacher also listens for whether any student spontaneously uses the word 'protein' — if so, note the student's name to leverage in Phase 3.

	Groups record their reasoning in a two-column table: 'What the evidence shows us' vs 'What we still need to find out.'	Debrief cold-calls — 5 groups report one item from each column: 'Group 3 — Fatuma, tell us one thing the evidence SHOWS.' [WAIT 10 seconds.] After each response: 'Does anyone want to add to or challenge that?' [WAIT 8 seconds.] After all groups report, say: 'Notice that almost every group wrote something like: we don't yet know what specific chemicals are different between ugali and githeri. That is EXACTLY the question that this whole unit will answer. Hold onto that question — it belongs on our Driving Question Board, which we will build next.'		
Explain Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Eukaryotes to Multicellular Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a brief, structured whole-class activity to introduce — at a conceptual level only — the six major chemicals of life: carbohydrates, proteins, lipids (fats and oils), vitamins, mineral salts, and water. This is NOT a lecture; it is a guided inquiry using a 'Chemical Families Card Sort.' Each group receives a set of 12 cards: 6 'Chemical Family' cards (one per chemical class, with a simple name and one-line description) and 6 'Kenyan Food Example' cards (ugali, githeri, sukuma wiki, nyama choma, fish from Lake Victoria, mandazi). Groups match each food to the chemical family they think it is RICHEST in, and then place the cards on a large sheet of paper, drawing arrows and writing reasons. Groups then do a gallery walk — they place their sorted cards on a desk and walk to see other groups' sorts, using sticky	Distribute the Chemical Families Card Sort. Say: 'Each group has 6 chemical family cards and 6 Kenyan food cards. Your task: match each food to the chemical family you think it is RICHEST in. You may not use a phone or textbook yet — use only what you already know or can reason out. You have 5 minutes.' [WAIT 5 minutes.] Circulate and note interesting matches and mismatches (e.g. students who put ugali under 'protein' or mandazi under 'vitamins'). Do NOT correct — say: 'Interesting reasoning — make sure you write your reason on the card.' Gallery Walk instructions: 'Stand up. Leave your sorted cards on your desk. You will now rotate around the room. At each group's	Strategy: Card Sort + Gallery Walk with Peer Annotation. Card sorts are a high-leverage classification strategy (NGSS SEP 6) that force students to articulate criteria for grouping. Using Kenyan food examples (githeri, sukuma wiki, nyama choma, fish from Lake Victoria) ensures the classification task is grounded in the phenomenon and culturally meaningful. The gallery walk transforms the activity into a peer-review process, exposing students to multiple reasoning strategies and building the metacognitive habit of comparing their thinking to others'. The sticky-note protocol ('I agree because / I wonder if') scaffolds constructive scientific discourse.	Observable indicator: Each group's card sort sheet shows written reasoning (not just matches) for at least 4 of the 6 food cards. Teacher specifically checks whether githeri is linked to protein (correct) and ugali to carbohydrates (correct) — these two matches are the conceptual core of the phenomenon. If most groups have both correct, the teacher can fast-track in Phase 4; if most have them wrong, the teacher flags this explicitly: 'This is exactly what our chemical tests next lesson will help us resolve.' Teacher also records which chemical families (e.g. vitamins, mineral salts) are the most confused — these will receive extra emphasis in Lessons 4–5.

	notes to leave one 'I agree because…' and one 'I wonder if…' comment on each group's display.	desk, read their sort, then leave one sticky note that says I AGREE BECAUSE… and one that says I WONDER IF… You have 6 minutes for the gallery walk.' [TIMER: 6 minutes.] After gallery walk, groups return to their own desks. 'Look at the sticky notes left on your sort. Does anyone want to change a match based on what you saw? You have 90 seconds to revise.' [WAIT 90 seconds.] Whole-class debrief: cold-call 3 groups to share their most debated match. 'Otieno — which match caused the most disagreement in your group and why?' [WAIT 12 seconds.] After each response, ask: 'Did another group sort that food differently? Wanjiku — what did your group decide?' [WAIT 10 seconds.] Close with: 'You have just built a first-draft map of the chemicals of life. In the next lessons, we will use chemical tests to find out which of your matches are correct — and some of your predictions WILL be wrong, and that is perfectly good science.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Dehydration synthesis or a condensation reaction] Source: Khan Academy (English - US curriculum)	Each student receives three sticky notes (different colours: pink = question I have, yellow = something I think I know, blue = something I want to test or investigate). Students individually write one entry per sticky note in response to the phenomenon. The teacher then facilitates a whole-class DQB construction on a large display board at the front of the	Distribute three sticky notes per student. Say: 'Pink note: write one QUESTION you have after everything we have done today. Yellow note: write one thing you now THINK YOU KNOW about the chemicals of life. Blue note: write one thing you want to TEST or INVESTIGATE. You have 3 minutes. Write in full sentences.' [WAIT 3 minutes — circulate,	Strategy: Three-Colour Sticky Note DQB Protocol. The three-colour protocol (questions / knowledge / investigations) structures the DQB so it tracks three distinct dimensions of scientific thinking simultaneously: curiosity (what I want to know), current knowledge (what I think I know), and epistemic agency (what I want to DO to find out). Clustering the	Observable indicator: Every student has placed all three sticky notes on the DQB within the allocated time. Teacher photographs the board at the end of the lesson for a record. Teacher reviews the pink (question) notes after class to identify: (1) the most common misconceptions embedded in questions (e.g. 'Is protein only in meat?' — a very

🔍 [Search ARES for similar videos] 📖 **HTML:** [Evolution of Eukaryotes to Multicellular Life (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	classroom. Students come up and place their sticky notes, guided by the teacher to cluster them into emerging themes: What is food made of? What does each chemical do? Why does the body look different without a chemical? How can we test which chemicals are in food? The board is headed by the driving question: 'Why do two children in Kisumu eating the same amount of food every day end up with completely different bodies — and what does that tell us about what food is really made of?' The DQB will remain on display and be updated at the start and end of every subsequent lesson in the unit.	prompt quiet students with: 'What is the most confusing thing for you right now? Write that as a question.'] DQB construction: 'Come up row by row and place your notes on the board. As you place yours, read what is already there and try to put yours near a similar idea.' [Allow 4 minutes for this — create a gentle, purposeful atmosphere, not chaotic.] After all notes are placed, teacher reads selected entries aloud and clusters them live: 'I can see a cluster forming around the question of what chemicals ARE in food — look here. And another cluster around what those chemicals DO to the body. And a third cluster around HOW we could test for them. These three clusters are exactly the three big ideas this unit will answer.' Point to the driving question banner: 'Every lesson in this unit, we will come back to this board. We will move questions from UNANSWERED to ANSWERED as we do experiments and build knowledge. By the last lesson, I expect this board to look completely different — because your thinking will have grown.' Cold-call 3 students to read their pink question aloud: 'Akinyi — read your question.' [WAIT 5 seconds after reading.] Ask the class: 'Raise your hand if you have a similar question.' This validates individual curiosity as shared	notes into themes models how scientists organise questions into research programmes. Displaying the board permanently makes scientific progress visible and tangible — students can physically move or annotate notes as the unit develops, reinforcing the idea that science is a cumulative, revisable process.	common Kenyan student misconception that beans are not a protein source); (2) the most sophisticated questions that can anchor later lessons (e.g. 'If githeri has protein and ugali doesn't, why doesn't the sick child just eat githeri?'). These questions are re-written on larger cards by the teacher and added to the DQB before Lesson 2 to signal that student questions are taken seriously.

		inquiry.		
Model Building Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Eukaryotes to Multicellular Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students individually draw their first explanatory model in their science notebooks. The model must show: (a) ugali going into a body on one side, (b) githeri going into a body on the other side, (c) what the student thinks is happening inside each body based on their current understanding, and (d) why this leads to the two different health outcomes seen in the Kisumu children. Students use annotated diagrams with arrows — they do not need to know all the correct chemical names yet; they use whatever language they have. Below the model, students write a 'Model Limitations Statement': 'My model cannot yet explain _______ because _______.' This forces metacognitive awareness of what the model does and does not account for. Students share their model with one partner and give each other one 'warm' comment (something the model explains well) and one 'wonder' comment (something the model doesn't explain yet). Models are kept in science notebooks and will be revised at the end of every lesson in the unit.	Say: 'For the last activity today, I want you to build your first explanatory model. A model in science is a diagram that shows your BEST CURRENT EXPLANATION for something. Yours will not be perfect — that is fine. In fact, the whole point of this unit is to improve this model lesson by lesson.' Display the model prompt on the board: 'Draw: ugali → body | githeri → body. Show what you think is happening inside each body. Show why the health outcomes are different. Label everything. Use arrows to show cause and effect. You have 6 minutes.' [WAIT 6 minutes — circulate and quietly affirm effort, not correctness: 'I can see you are thinking hard about the arrows — keep going.'] After 6 minutes: 'Below your model, write this sentence and complete it: My model cannot yet explain _______ because _______. You have 90 seconds.' [WAIT 90 seconds.] Peer share: 'Turn to your partner. Share your model. Partner A gives one WARM comment — something the model explains well. Partner B gives one WONDER comment — something the model doesn't explain yet. Then swap. You have 3 minutes.' [WAIT 3 minutes.] Close the lesson: 'Put a box around your model and write	Strategy: Initial Explanatory Model + Model Limitations Statement. Initial model building is a core NGSS Science and Engineering Practice (SEP 2: Developing and Using Models). Requiring students to articulate what their model CANNOT explain is a research-backed metacognitive scaffold — it prevents premature closure and keeps students in a productive 'epistemic discomfort' that motivates further inquiry. The warm/wonder peer protocol mirrors the scientific practice of peer review and builds collaborative discourse norms early in the unit. Preserving the initial model (no erasure) creates a physical artefact of conceptual change that students can reference throughout the unit, reinforcing that revision is the engine of scientific progress.	Observable indicator: Every student's model includes at least (a) both foods labelled, (b) at least one arrow showing a cause-effect relationship, and (c) a completed 'Model Limitations Statement.' Teacher collects science notebooks at the end of the lesson (or photographs open pages) and reviews the models for the following specific diagnostic information: (1) Does the student connect the missing chemical in ugali to the physical symptoms (even vaguely)? (2) Does the student use any chemical vocabulary (starch, protein, vitamins) or does the model rely entirely on general terms ('bad food', 'less nutrition')? (3) Is the 'Model Limitations Statement' substantive (e.g. 'My model cannot yet explain why the belly swells, not the whole body') or superficial ('I don't know')? This information is used to differentiate instruction in Lesson 2.

		today's date. Do not erase anything — even if you think it is wrong. These models belong to you. By the last lesson of this unit, you will draw a new model next to this one, and you will be amazed at how much your thinking has grown. That growth IS the science.' Final closing statement: 'Today you did what scientists at KEMRI and nutritionists working with Kenyan families do every day — you looked at a real problem in your community, asked a question, and started building an explanation. You didn't just learn ABOUT science — you DID science.'		

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions in your professional journal:

1. PHENOMENON LAUNCH: Did the photographs of the two Kisumu children generate genuine curiosity and emotional engagement, or did students seem distant from the phenomenon? If students seemed disengaged, consider whether a more locally specific story (e.g. a named child from a specific Kisumu sub-county, or a short video clip) might anchor the phenomenon more powerfully in Lesson 1 of your next delivery.

2. PRIOR KNOWLEDGE RANGE: When you reviewed students' predictive diagrams and initial models, what was the range of prior knowledge in the class? Were there students who already knew the six chemical classes well (perhaps from primary school science)? If so, plan to deploy these students as 'expert consultants' during the card sort in future lessons rather than having them sit through content they already know.

3. DQB QUALITY: Were the pink (question) sticky notes substantive scientific questions, or were they mostly vague ('I want to know about food')? If questions were vague, in Lesson 2 model the difference between a 'big question' and a 'testable question' before students add to the DQB, using examples from today's board.

4. MISCONCEPTION ALERT: The most common misconception in Kenyan classrooms for this topic is that protein comes ONLY from animal sources (meat, eggs, milk) and that beans and githeri are not 'protein foods.' If you observed this misconception strongly in the card sort, prioritise addressing it explicitly at the start of Lesson 3 using the iodine and Biuret test results on githeri as direct counter-evidence.

5. SENSITIVITY CHECK: Did any student show signs of distress, hunger, or personal identification with the children in the photographs? Note any such students and consult the school counsellor if appropriate. Food insecurity is a real experience for some learners in Kenyan schools, and the phenomenon must always be handled with care and dignity.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed two photographs of children from Kisumu — one with signs of kwashiorkor (swollen belly, thin limbs, discoloured hair, fatigue) and one healthy and energetic — and were told both children eat similar quantities of food daily, with Child A eating ugali and Child B eating githeri.
What did I learn?	Students learned that food is not simply 'food' — it is composed of different chemical substances (carbohydrates, proteins, lipids, vitamins, mineral salts, water), and that two foods can look similar in quantity but be chemically very different. They were introduced to the names of the six major chemical families of life and matched them to familiar Kenyan foods in a first-draft card sort.
How does this explain the phenomenon?	Students cannot yet fully explain why the two children have different bodies, because they do not yet know which specific chemicals are present in ugali versus githeri, what each chemical does in the body, or how to test for their presence. Their first explanatory models show a beginning of cause-effect thinking (missing something in ugali → sick body) but lack the chemical detail that food tests in upcoming lessons will provide. The driving question remains open: Why do two children in Kisumu eating the same amount of food every day end up with completely different bodies — and what does that tell us about what food is really made of?

LESSON 2 (80 minutes): What Is Food Really Made Of? Introducing the Six Chemicals of Life	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To introduce learners to the six major chemicals of life — carbohydrates, proteins, lipids, nucleic acids, vitamins/minerals, and water — and connect each chemical to the observable health differences seen in the two Kisumu children in the anchoring phenomenon.
Knowledge	Learners will identify and describe the six major chemicals of life (carbohydrates, proteins, lipids, nucleic acids, vitamins/minerals, water), state the general role of each chemical in living organisms, and explain why eating the same quantity of food does not guarantee the same health outcome if the chemical composition of those foods differs.
Skills	Learners will practise scientific prediction, analyse photographs as evidence, construct a class Driving Question Board (DQB), categorise foods by their dominant chemical components, and begin building a cumulative molecular model of food composition.
Attitudes	Learners will appreciate that science can explain everyday health inequalities observed in Kenyan communities, show curiosity about the molecular nature of food, and demonstrate respect for diverse initial ideas during class discussion.
Key Inquiry Question	What are the chemicals of life and what roles do they play in living organisms?
Purpose in Storyline	This is Lesson 2 of 6 in the evidence-gathering sequence. In Lesson 1 learners were introduced to the anchoring phenomenon (the two Kisumu children) and made initial observations. In this lesson learners move from surface-level observation ('they eat different foods') toward a chemical-level explanation by mapping the six chemicals of life onto the foods each child eats. This lesson establishes the conceptual framework — the six chemicals — that every subsequent lesson will investigate in depth through food tests and body-function analysis. The DQB initiated here becomes the living record of the class's growing understanding across all six lessons.
Safety Notes	No hazardous chemicals are used in this lesson. Photographs of children with kwashiorkor must be handled with sensitivity; remind learners that these are real conditions affecting real families and that the purpose of studying them is to understand and prevent suffering. Learners must not use stigmatising language. If any learner shows signs of personal distress when viewing the photographs, the teacher should respond with empathy and, if necessary, refer the learner to the school counsellor.

B. LESSON OVERVIEW
Learners begin by revisiting the anchoring phenomenon — photographs of the child eating ugali (maize starch only) who has developed kwashiorkor, and the healthy child eating githeri (maize and beans). Working in pairs, learners write their best prediction about what food is made of at the chemical level and why quantity alone cannot explain the difference in health outcomes. The class shares predictions and the teacher initiates the Driving Question Board. The teacher then guides learners through an interactive introduction to the six major chemicals of life using locally sourced food cards (ugali, sukuma wiki, omena, avocado, beans, milk, mango). Learners use a graphic organiser to record the name, key role, and a Kenyan food source for each chemical. The lesson closes with learners revising their initial predictions and adding new 'wonderings' to the DQB, and each learner sketching a first-draft molecular model of ugali versus githeri to explain the phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners are seated in pairs. Each	Display the two photographs on	Strategy: Prediction Anchoring. By	Observable indicator: Each learner

📹 **VIDEO:** [Dehydration synthesis or a condensation reaction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Evolution of Eukaryotes to Multicellular Life (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	pair receives a printed A5 card showing two side-by-side photographs: (1) the child from Kisumu with kwashiorkor — swollen belly, thin limbs, reddish discoloured hair; (2) the healthy child eating githeri. Below the photographs the card reads: 'Both children eat roughly the same amount of food every day. Child A eats ugali. Child B eats githeri. What do you think food is actually made of at the chemical level? Why do you think the same quantity of different foods causes such different results in the body?' Learners write individual predictions in their science notebooks (3 minutes silent writing), then discuss with their partner (2 minutes). Each pair agrees on one joint prediction sentence to share with the class.	the board (projected or printed poster). Say verbatim: 'Before we learn anything new today, I want to know what YOU already think. Look at these two children — both in Kisumu, both eating food every day. In your notebook, write your best scientific guess: what do you think food is actually made of at the chemical level, and why does the type of food — not just the amount — seem to matter so much?' Allow 3 minutes of silent individual writing. Do NOT answer questions during this time — redirect with: 'Trust your thinking — write what you believe right now.' After 3 minutes, say: 'Now share your prediction with your partner. You have 2 minutes. Agree on one sentence you both believe.' After pair discussion, cold-call 4 pairs (select a mix of confidence levels — do not only pick raised hands). For each, ask: 'What is your joint prediction, and what experience or knowledge are you drawing on?' Allow 10 seconds of WAIT TIME after each response before responding or moving to the next pair. Record key words from predictions on the board in a 'Before' column without correcting — affirm scientific thinking: 'That is exactly the kind of question a biochemist would ask.'	committing predictions to writing before instruction, learners activate prior knowledge and create a cognitive 'gap' between what they think they know and what they are about to discover. This gap drives motivation and primes learners to notice evidence that confirms or challenges their prediction — a core science practice (NGSS SEP 6: Constructing Explanations).	has written at least one full prediction sentence in their notebook before pair discussion begins. Teacher circulates and reads over shoulders — look for whether learners mention any chemical or nutrient terms (starch, protein, vitamins) or whether they frame the difference purely in quantity terms ('githeri has more food'). This reveals the starting point for conceptual development. Note 2–3 common misconceptions to address explicitly during the Explain phase.
Observe Phase 📹 **VIDEO:** [Dehydration synthesis or a condensation reaction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners participate in a structured 'Food Card Sort' activity. Each pair receives a set of 10 laminated food cards featuring Kenyan foods with photographs: ugali, githeri, omena (small dried fish from Lake Victoria), sukuma wiki, avocado, milk, mango, groundnuts, eggs, and white sugar. On the reverse of each card is the food's approximate composition by	Distribute food card sets to each pair. Say: 'Your job right now is not to get the right answer — your job is to OBSERVE patterns. Look at these foods. Sort them into groups based on whatever pattern you notice in the composition numbers on the back. You have 5 minutes.' Circulate and observe without correcting. After 5 minutes, project the Six Chemicals of Life reference	Strategy: Structured Card Sort with Progressive Reveal. The initial open sort activates learners' pattern-recognition skills without imposing categories, honouring learner agency (NGSS SEP 8: Obtaining, Evaluating and Communicating Information). The progressive reveal of the scientific framework then gives learners the vocabulary to name patterns they	Observable indicator: After the second sort, each pair's graphic organiser has all six chemicals named with at least one correct Kenyan food example per row. Teacher checks 6 pairs' organiser sheets during circulation — specifically look for whether learners correctly distinguish proteins from carbohydrates (common confusion point: githeri is

📖 **HTML:** [Evolution of Eukaryotes to Multicellular Life (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	percentage (e.g., ugali: ~75% carbohydrate, ~2% protein, ~1% fat, ~22% water). Learners are asked to sort the cards into groups based on any pattern they observe — they are NOT yet told the categories. After sorting (5 minutes), pairs examine a reference chart projected on the board: a simple table showing the six chemicals of life with a one-line description and one example each. Learners then re-sort their food cards to match each chemical category, and record in their graphic organiser which Kenyan foods are richest in each chemical.	chart. Say verbatim: 'Scientists have discovered that all living things are built from six major groups of chemicals. Let me show you the framework biochemists use.' Read each row aloud, pausing after each chemical to ask the class a quick question — cold-call 2 learners per chemical (12 cold-calls total): 'Looking at your food cards — which food do you think is highest in [chemical name]? Take 10 seconds.' Allow 10 seconds WAIT TIME. After the chart is reviewed, say: 'Now re-sort your food cards using THIS framework as your guide. Update your graphic organiser — write the chemical name, its main role in the body, and one Kenyan food example for each row.'	already noticed, creating a bridge between intuitive and formal scientific knowledge. Using Kenyan foods makes the abstract chemistry personally relevant and memorable.	often perceived as 'just a starch food'). Ask targeted probing question to 3 pairs: 'You placed beans under protein — what is it about beans that makes them a protein source rather than a carbohydrate source?'
Explain Phase 📹 **VIDEO:** [Dehydration synthesis or a condensation reaction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Evolution of Eukaryotes to Multicellular Life (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class explanation of each of the six chemicals of life, explicitly connecting each one back to the two Kisumu children. Learners follow along using their graphic organiser, adding detail to each row. A key focus is on proteins: what they are made of (amino acids), what they do (build and repair body tissues — muscles, enzymes, hormones), and what happens when the body receives insufficient protein despite adequate calories — kwashiorkor. Learners then complete a 'Cause and Effect' structured writing task: in 3–4 sentences, they explain in their own words why the child eating ugali alone developed kwashiorkor, using the names and roles of at least three chemicals of life in their explanation.	Begin with carbohydrates. Say: 'Ugali is almost entirely carbohydrate — specifically starch. Starch gives the body energy. So why is the child eating ugali sick? Let's think about that.' Pause 10 seconds. Cold-call 2 learners. Then work through each chemical systematically — for each one, use the sentence frame: 'The chemical is [name]. Its job in the body is [role]. If the body doesn't get enough, [consequence].' For proteins, pause for extended explanation: 'Proteins are the BUILDING chemicals. Every muscle fibre in your body, every enzyme that digests your food, every hormone that coordinates your body — all made of protein. When the Kisumu child eats only ugali, they get plenty of energy — but almost zero protein. The body cannot build or repair tissues. The swollen belly, the thin muscles, the changed hair colour — these are	Strategy: Phenomenon-Anchored Direct Instruction with Embedded Socratic Questioning. Rather than presenting the six chemicals as abstract facts, the teacher anchors each chemical to observable evidence from the phenomenon — the swollen belly (protein deficiency), the energy from ugali (carbohydrates), the healthy skin of the githeri child (protein + micronutrients). Socratic cold-calling throughout ensures learners are actively constructing meaning rather than passively receiving information (NGSS SEP 6: Constructing Explanations and Designing Solutions).	Observable indicator: Learners' Cause and Effect written responses correctly name at least three chemicals of life, assign each a specific role, and connect the absence of one chemical (protein) to at least two visible symptoms in the kwashiorkor child. Teacher collects 8 written responses (selected randomly) to read between lessons — use these to identify learners who still conflate energy foods with building foods, and plan a brief re-teach at the start of Lesson 3.

		the body's cry for protein.' Show a simple diagram of amino acids linking into a protein chain on the board. For lipids, connect to avocado and omena. For vitamins and minerals, reference sukuma wiki. For water, ask: 'Why do you think water is listed as a chemical of life?' — 10 seconds WAIT TIME, cold-call 3 learners. For nucleic acids, say: 'DNA and RNA — we will explore these in depth later in the unit, but know that every cell in both children's bodies contains nucleic acids carrying the instructions for life.' Close the explanation phase by returning to the phenomenon: 'So now — using what we have just learned — turn to your partner and spend 2 minutes: which of the six chemicals is MOST responsible for the difference between these two children, and how do you know?' Cold-call 3 pairs to share. Then direct learners: 'Write your 3–4 sentence Cause and Effect explanation now. Use the names of at least three chemicals of life.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Dehydration synthesis or a condensation reaction]() Source: Khan Academy (English - US curriculum) [🔍 Search ARES for similar videos]() 📖 **HTML:** [Evolution of Eukaryotes to Multicellular Life]()	The class creates the unit's Driving Question Board together. The central driving question is displayed at the top: 'Why do two children in Kisumu eating the same amount of food every day end up with completely different bodies — and what does that tell us about what food is really made of?' Each learner receives two sticky notes. On the first (green): they write one thing they now understand that they did not understand at the start of the lesson. On the second (yellow): they write one new question the lesson has sparked — something they are now curious about or	Prepare the DQB on a large section of classroom wall or on a projected shared document. The central driving question should be written in large font at the top. Distribute sticky notes. Say: 'This board is going to grow across all six lessons. It is our record of how our thinking is changing. Right now, write ONE thing you understand better than you did an hour ago — green note. And ONE new question this lesson has given you — yellow note.' Allow 3 minutes for silent writing. Then direct learners to post notes. Once all notes are posted, read 5–6 yellow notes aloud without	Strategy: Collaborative Driving Question Board as a Living Knowledge Map. The DQB externalises the class's collective thinking, making the progression from confusion to understanding visible over time. It validates learners' questions as the engine of scientific inquiry (NGSS SEP 1: Asking Questions and Defining Problems) and creates accountability — learners can see their own question answered across the unit, giving science learning a narrative arc rooted in genuine curiosity.	Observable indicator: Every learner has posted one green and one yellow sticky note. Teacher reads all green notes to check for conceptual accuracy — if a green note contains a misconception (e.g., 'I now understand that ugali has no nutrients at all'), the teacher addresses it briefly and supportively at the start of the next lesson rather than publicly correcting in the moment. Yellow notes are catalogued by the teacher as a diagnostic of curiosity and remaining confusion; a minimum of 60% of yellow notes should reference chemical-level questions (not just 'what is the

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	confused by. Learners post their notes on the class DQB under two columns: 'What We Now Understand' and 'What We Still Wonder'. The teacher reads 5–6 yellow notes aloud and groups similar questions into themes (e.g., 'How do we actually TEST for these chemicals?' — pointing forward to Lesson 3's food tests; 'Does the body make any of these chemicals itself?').	identifying the author. Say: 'I notice several of you are asking HOW we can detect these chemicals in food. That is exactly what a biochemist would ask — and it is exactly what we are going to do in our next lesson.' Group the yellow notes visually into 3–4 question clusters. Say: 'These clusters are going to guide our investigation for the rest of this unit. Each lesson, we will come back to this board and move questions from yellow to green as our understanding grows.' Take a photograph of the board to document its state at Lesson 2.		right answer') to indicate genuine scientific curiosity is developing.
Model Building Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Eukaryotes to Multicellular Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners draw a first-draft 'molecular model' of each child's food in their science notebook. The model does not need to be chemically accurate — it is a diagrammatic representation. Learners draw two circles side by side: one labelled 'Ugali (Child A)' and one labelled 'Githeri (Child B)'. Inside each circle, they use a simple pie-chart or icon-based representation to show which of the six chemicals of life are present in significant amounts in each food. They annotate each chemical with a brief role note and, where relevant, a consequence of deficiency. At the bottom of the model, learners write one sentence connecting their model to the driving question. Pairs compare their models and discuss: 'What is the biggest chemical difference between these two foods, and how does that explain what we see in the photographs?'	Say: 'Scientists do not just describe the world in words — they build models that show relationships. I want you to draw your first model of what is actually inside these two foods at the chemical level. Use the six chemicals of life as your key. Show me — visually — why these two foods produce such different bodies.' Allow 7 minutes for individual model drawing. Circulate and give specific feedback: 'I can see you have carbohydrates taking up most of the ugali circle — good. Now show me protein — how much protein is in ugali compared to githeri?' After 7 minutes, say: 'Compare your model with your partner's. Where do you disagree? Spend 2 minutes discussing — whose model better explains the photographs, and why?' Cold-call 2 pairs to share their comparison. Say: 'These models are YOUR starting point. Every lesson we do, you are going to revise this model and make it more accurate and more detailed. By Lesson 6, your model will be the explanation of	Strategy: Progressive Model Revision (Modelling as a Science Practice). Beginning with an imprecise, personally constructed model and iteratively revising it across lessons mirrors how scientists actually build understanding (NGSS SEP 2: Developing and Using Models). The deliberate instruction to preserve the first draft makes the growth in understanding visible and celebrates it — building scientific identity and self-efficacy. Anchoring the model to the two Kisumu children ensures the model is always in service of explaining a real, meaningful phenomenon rather than being an abstract exercise.	Observable indicator: Each learner's notebook contains a two-circle model with all six chemicals of life labelled in at least one of the two circles, and protein is shown as significantly more present in the githeri circle than the ugali circle. The closing sentence connects the chemical difference to a visible difference in the children's bodies. Teacher photographs 4–5 models to use as discussion anchors at the start of Lesson 3. Look specifically for learners who are still not distinguishing between energy chemicals (carbohydrates, lipids) and structural/functional chemicals (proteins, nucleic acids) — these learners need targeted scaffolding in Lesson 3.

		the entire phenomenon. Keep this page — do NOT erase it. Your first draft is evidence of your scientific thinking growing.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. PREDICTION QUALITY: What did the predictions learners made in Phase 1 reveal about their prior knowledge of food chemistry? Were learners thinking at a molecular level or purely at a quantity/type level? How will this shape your approach in Lesson 3?

2. COLD-CALLING PATTERNS: Did you successfully cold-call learners across different confidence levels, gender, and seating positions? Did any group of learners remain consistently silent? Plan a strategy to draw those learners in during Lesson 3.

3. PHENOMENON CONNECTION: At the close of the lesson, could learners clearly articulate — in their own words — why the ugali-eating child developed kwashiorkor? If not, which phase broke down, and how will you re-teach?

4. DQB QUALITY: Were the yellow sticky note questions chemical-level questions (evidence of genuine scientific curiosity) or procedural questions ('Will this be on the exam?')? If mostly procedural, consider how to cultivate a classroom culture of curiosity more intentionally in Lesson 3.

5. MODEL SOPHISTICATION: Did learners' first-draft models show the six chemicals, or did learners revert to drawing food items (ugali, githeri) rather than chemical components? If so, the distinction between food and the chemicals IN food needs reinforcing at the start of Lesson 3.

6. KWASHIORKOR SENSITIVITY: Were there any signs of emotional distress when learners viewed the photographs? Was the classroom discussion conducted with dignity and appropriate scientific framing? Adjust the opening of Lesson 3 if the emotional register needs recalibration.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed two photographs of children in Kisumu — one with kwashiorkor (eating ugali only) and one healthy (eating githeri) — and sorted laminated cards of 10 Kenyan foods (ugali, githeri, omena, sukuma wiki, avocado, milk, mango, groundnuts, eggs, white sugar) by chemical composition using percentage data printed on the reverse of each card.
What did I learn?	Learners learned that all food is composed of six major chemicals of life — carbohydrates, proteins, lipids, nucleic acids, vitamins/minerals, and water — each with a distinct role in the body. They learned that ugali is almost entirely carbohydrate (energy) with very little protein (building/repair), while githeri combines carbohydrate (maize) with significant protein (beans), explaining why the same quantity of food produces dramatically different health outcomes.
How does this explain the phenomenon?	Learners explained, through a written Cause and Effect response and a first-draft molecular model, why the child eating ugali alone developed kwashiorkor: the body received sufficient energy (carbohydrates) but insufficient protein, so it could not build or repair muscle tissue, produce enzymes and hormones, or maintain normal fluid balance — causing the visible symptoms of swollen belly, thin limbs, discoloured hair, and fatigue seen in the photograph.

LESSON 3 (80 minutes): Observe (Part 2): Lipids and Vitamins — Structure, Function, and Food Tests	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the structure and function of lipids and vitamins as chemicals of life, and to perform food tests to identify their presence in common Kenyan foods.
Knowledge	By the end of the lesson, learners should be able to: (a) describe the chemical structure of lipids (glycerol and fatty acids) and classify vitamins (fat-soluble and water-soluble); (b) explain the roles of lipids and vitamins in living organisms; (c) identify lipids in food samples using the ethanol emulsion test and translucent spot test.
Skills	By the end of the lesson, learners should be able to: (a) perform the ethanol emulsion test and the translucent spot (grease-spot) test to identify lipids in food; (b) record, interpret, and communicate experimental observations accurately; (c) link experimental evidence to the anchoring phenomenon.
Attitudes	By the end of the lesson, learners should be able to: (a) appreciate the importance of dietary diversity in maintaining health; (b) show curiosity and persistence during practical investigations; (c) value accuracy and honesty in recording scientific observations.
Key Inquiry Question	What are the chemicals of life and what roles do they play in living organisms? How can we identify the chemicals present in different foods using chemical tests?
Purpose in Storyline	In Lesson 2, learners gathered evidence about carbohydrates (starch and simple sugars) and proteins — establishing that ugali provides carbohydrates but lacks sufficient protein, explaining the Kisumu child's kwashiorkor. In this lesson (Lesson 3), learners extend their chemical investigation to lipids and vitamins. By testing fat-containing Kenyan foods (avocado, groundnuts, milk, coconut), they discover that lipids are energy stores and essential membrane components, and that vitamins — though needed in tiny amounts — are critical for metabolic processes and immunity. This deepens learners' understanding of why dietary diversity, not just food quantity, determines health outcomes, advancing the driving question toward a complete chemical model of food.
Safety Notes	Caution: Ethanol is flammable — ensure NO open flames during the ethanol emulsion test. Learners must not taste any food samples used in experiments. Dispose of used ethanol in the designated waste container. Wear safety goggles when handling ethanol. Learners with nut allergies must not handle groundnut samples directly — assign an alternative food (avocado) and inform the teacher in advance. Wash hands thoroughly after handling all food and chemical samples.

B. LESSON OVERVIEW

Learners begin by predicting which Kenyan foods contain fats, drawing on prior knowledge from Lesson 2. They then perform two lipid food tests — the ethanol emulsion test and the translucent (grease-spot) test — on a range of local foods (groundnuts, avocado, ugali, githeri, coconut flesh, whole milk). Through structured observation and data recording, learners gather evidence that lipids are concentrated in certain foods absent from the Kisumu child's ugali-only diet. A guided reading activity on vitamins (fat-soluble vs. water-soluble) connects lipid consumption to vitamin A and D absorption. Learners revisit the anchoring phenomenon, updating their Driving Question Board with new chemical evidence. The lesson closes with a model-building revision where groups add lipids and vitamins to their cumulative chemical map of food.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Endomembrane system Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Meet the Family: Inv..:Segment 6 Single Letter Codes and DNA Codons for Amino Acids Source: MIT Blossoms 🔍 Search ARES for similar readings	Learners are shown a visual display of eight common Kenyan foods: ugali, groundnuts, avocado, githeri, coconut flesh, whole milk, sukuma wiki, and mandazi. Working individually (2 minutes), each learner writes a prediction on a sticky note: 'I think the foods that contain the MOST fat are ___ because ___.' Learners then share predictions with a shoulder partner (1 minute). The teacher cold-calls 4 pairs to share. Learners record their predictions in their science notebooks under the heading: 'My Lipid Prediction — Lesson 3.' The teacher anchors the activity back to the phenomenon: 'Remember our child in Kisumu eating only ugali. We found it has carbohydrates but little protein. Today we ask — does it have fats? And does that matter?' Learners vote with thumbs up/down/sideways on whether they think fat deficiency could also affect the Kisumu child.	1. Display the food images prominently on the board or screen. Say: 'Look at these eight foods you know from your homes and markets. Before we touch anything, I want your brain to make a guess — not a random guess, a reasoned one.' 2. Give explicit wait time: 'You have 2 minutes of silent thinking time. Write your prediction — no talking yet.' [WAIT 2 minutes.] 3. Say: 'Now turn to your shoulder partner. Share your prediction and your reason. You each have 30 seconds.' [WAIT 1 minute.] 4. Cold-call exactly 4 pairs: 'Pair at the window — what did you predict and why?' Probe each with: 'What evidence from your daily life made you say that?' [WAIT 10 seconds after each probe.] 5. Record 3–4 student predictions verbatim on the board under 'Our Predictions.' Say: 'We are scientists today — we don't throw away our predictions, we TEST them.' 6. Connect to phenomenon: 'The child in Kisumu eats ugali daily. We already know ugali is mostly starch, low in protein. Our question today: what about fats? Could that gap also hurt the body?' [WAIT 15 seconds for silent reflection.] 7. Conduct thumbs vote on fat deficiency impact. Acknowledge all responses without confirming: 'Interesting — we will find out.'	Strategy: Anchored Prediction with Personal Evidence. Learners activate prior knowledge about familiar foods before instruction, reducing cognitive load when experimental data arrives. By requiring a reason ('because ___'), the teacher surfaces existing mental models to be refined or challenged during observation. Connecting predictions to the phenomenon ensures the activity is not isolated trivia but drives the unit's central question forward.	Observable indicator: Each learner has a written prediction with a reason in their notebook before observation begins. Teacher circulates during silent thinking time and scans notebooks — look for learners who list only one food (prompt: 'Can you think of a food your family cooks in oil or fat?') and learners who cannot give a reason (prompt: 'Think about what ugali feels like vs. groundnuts — what is different?'). Cold-call responses reveal whether learners conflate 'fatty foods' with 'unhealthy foods' — note this misconception for targeted correction during Phase 3.
Observe Phase 📹 **VIDEO:** Endomembrane system Source: Khan Academy (English	Learners work in groups of 4–5 at lab stations. Each station has labelled food samples: (A) ugali paste, (B) mashed groundnuts, (C) mashed avocado, (D) githeri paste, (E) coconut flesh (grated), (F) whole milk, (G) sukuma wiki	1. Before learners begin, demonstrate the translucent spot test using ONE sample (coconut flesh) at the front: 'Watch what I do — I rub, I wait, I observe. I am not telling you the result yet — you will discover it.' 2. Distribute lab	Strategy: Structured Data Collection with Parallel Tests. Using TWO independent tests for lipids (spot test and ethanol emulsion) models authentic scientific practice — scientists rarely rely on a single test. The	Observable indicators: (1) Results table is complete for all 8 food samples across both tests — teacher checks for consistency between the two test columns (if a food tests positive in one and negative in the other, prompt the

- US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Meet the Family: Inv..:Segment 6 Single Letter Codes and DNA Codons for Amino Acids Source: MIT Blossoms 🔍 Search ARES for similar readings	(blended), (H) mandazi. ACTIVITY 1 — Translucent Spot Test (10 min): Each group rubs a small amount of each food sample onto a piece of brown paper bag, then holds it up to light after 5 minutes. Lipid-containing samples leave a permanent translucent (greasy) spot that does NOT disappear when dry; non-fat samples leave a spot that dries and disappears. Learners record results in a results table (Food Sample / Spot Present? / Spot Permanent? / Contains Lipid?). ACTIVITY 2 — Ethanol Emulsion Test (12 min): For each food sample, learners: (i) crush/dissolve a small amount in 2 cm^3 ethanol in a test tube; (ii) filter or allow solids to settle; (iii) pour the clear ethanol layer into a test tube of distilled water. A cloudy white emulsion = lipid present; clear solution = lipid absent. Learners record results in the same table. ACTIVITY 3 — Guided Reading on Vitamins (8 min): Groups read a one-page structured text (provided by teacher) titled 'Vitamins: The Micro-Managers of Your Body,' covering fat-soluble vitamins (A, D, E, K) and water-soluble vitamins (B-complex, C), their food sources (local Kenyan examples), and deficiency diseases (night blindness, rickets, scurvy, beriberi). Each learner annotates: underline the role, circle the deficiency disease, star the Kenyan food source. Groups compile a shared 'Vitamin Summary Strip' poster strip (3 columns: Vitamin / Role / Kenyan Food Source). SHARE OUT: Each group nominates a 'data reporter' who shares one surprising finding	instructions and results tables. Say: 'Read the entire procedure BEFORE you touch anything. You have 2 minutes to read silently.' [WAIT 2 minutes.] 3. Circulate to each group during Activity 1. At each station, ask: 'What are you actually observing — describe it in words, not just yes/no.' [WAIT 10 seconds.] Probe: 'Why does the spot need to be permanent? What would a water-based food leave behind?' [WAIT 15 seconds.] 4. Before Activity 2, issue the ethanol safety reminder verbatim: 'Ethanol is flammable. There are NO flames in this room while we use ethanol. Goggles on. If you spill, tell me immediately.' Scan room to confirm all goggles are worn before proceeding. 5. During Activity 2, cold-call 3 groups mid-activity: 'Tell me what you see in your test tube right now — not what you expected, what you SEE.' [WAIT 10 seconds per group.] 6. During Activity 3 (Guided Reading), circulate and check annotations. Prompt learners who are underlining everything: 'If you underline everything, nothing stands out — what is the ONE most important role of Vitamin A?' [WAIT 10 seconds.] 7. After share-out, say: 'Keep your results tables — we will need them in Phase 3 to build our argument about the Kisumu child.'	parallel structure allows learners to cross-validate their findings, building confidence in evidence. The guided reading on vitamins uses annotation (underline/circle/star) as a discipline-specific literacy strategy, requiring learners to process text actively rather than passively. The Vitamin Summary Strip creates a shared artefact that externalises group understanding and prepares for Phase 3 argumentation.	group to repeat or explain the discrepancy). (2) Learners correctly identify ugali as lipid-absent or very low in both tests — this is a critical data point for the phenomenon; if a group marks ugali as lipid-rich, intervene immediately. (3) Vitamin Summary Strip contains at least 4 vitamins with correct Kenyan food sources — look for common errors such as listing only 'vegetables' without specifics or confusing fat-soluble and water-soluble vitamins.

	with the class.			
Explain Phase 📹 **VIDEO:** Endomembrane system Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Meet the Family: Inv..:Segment 6 Single Letter Codes and DNA Codons for Amino Acids Source: MIT Blossoms 🔍 Search ARES for similar readings	Learners use their experimental data and the Vitamin Summary Strip to construct a written 'Evidence Paragraph' using the CER (Claim–Evidence–Reasoning) framework. The prompt is displayed on the board: 'CLAIM: The Kisumu child eating only ugali is likely deficient in [___]. EVIDENCE: From our food tests, we found that [___]. REASONING: This matters because lipids/vitamins do [___] in the body, and without them [___] could happen.' Groups draft their CER paragraph collaboratively (6 minutes), then each group shares their CLAIM sentence only with the class. The teacher records all claims on the board. A whole-class discussion follows (8 minutes): the teacher introduces the concept of essential fatty acids and fat-soluble vitamin absorption (Vitamins A, D, E, K are absorbed only in the presence of dietary fat), connecting the two investigations. Learners then individually write a 'So What?' sentence: 'This means that eating ugali alone, even in large quantities, will cause ___ because ___.' The lesson pauses for a 'Temperature Check' — the teacher poses the question: 'Could the child's eye problems (night blindness) and skin dryness ALSO be caused by the same dietary gap as the swollen belly?' Learners vote and justify to a partner before the teacher clarifies.	1. Display the CER prompt. Say: 'You have collected the evidence. Now you are going to use it like a scientist — to build an argument. A scientific argument has three parts: your Claim, your Evidence, and your Reasoning. I will model one sentence of each, then you continue.' Model briefly with a non-related example (e.g., 'Githeri contains protein — evidence: biuret test turned purple — reasoning: purple colour indicates peptide bonds…'). 2. Give groups 6 minutes for CER drafting. Circulate and listen — do NOT correct content yet, only prompt: 'Is your evidence from YOUR data table, or are you writing what you already believed?' [WAIT 10 seconds.] 3. Cold-call each group for their CLAIM sentence only. Record all claims verbatim on the board. Ask class: 'Which of these claims is supported by the most specific evidence from today's tests?' [WAIT 15 seconds.] 4. Introduce fat-soluble vitamin connection: 'Here is something surprising — Vitamins A, D, E, and K cannot be absorbed by your body without fat present. So if you eat no fat, even a vitamin-rich meal cannot fully protect you. Let that sink in.' [WAIT 15 seconds of silent reflection.] 5. Prompt individual 'So What?' sentence. Cold-call 3 individual learners (not group reporters) to share their sentences. Probe: 'What specific change in the child's body are you predicting — and which chemical is responsible?' [WAIT 10 seconds per learner.] 6. Temperature Check vote: Show of hands, then partner discussion (1 minute), then	Strategy: Claim–Evidence–Reasoning (CER) Framework. CER is a structured argumentation scaffold that mirrors how scientists communicate findings. By requiring learners to separate their claim (conclusion), evidence (data), and reasoning (mechanism/theory), the teacher disrupts the common student practice of restating observations as explanations. The Temperature Check on night blindness creates productive cognitive dissonance — learners realise the phenomenon (kwashiorkor child) may have MORE deficiency symptoms than initially visible, deepening engagement with the driving question. Recording all group claims on the board makes reasoning public and promotes peer critique.	Observable indicators: (1) CER paragraphs name a specific deficiency (e.g., Vitamin A, lipids) rather than vague terms like 'nutrients' — groups that write 'the child lacks nutrients' need probing: 'Which nutrient? Name the chemical class.' (2) Evidence sentences cite specific test results (e.g., 'ugali showed no permanent translucent spot and no cloudy emulsion') rather than general statements. (3) Individual 'So What?' sentences connect a missing chemical to a specific body function — this is the key transfer indicator; learners who cannot complete this sentence will need additional scaffolding in Lesson 4. (4) Temperature Check vote distribution — if fewer than 50% connect night blindness to fat deficiency, pause and revisit the fat-soluble vitamin concept before moving to Phase 4.

		reveal: 'Night blindness is caused by Vitamin A deficiency — and Vitamin A is fat-soluble. So yes, the SAME dietary gap (no fat, no varied food) causes MULTIPLE symptoms. The body is an interconnected system.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Endomembrane system Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Meet the Family: Inv..:Segment 6 Single Letter Codes and DNA Codons for Amino Acids Source: MIT Blossoms 🔍 Search ARES for similar readings	Each group revisits the class Driving Question Board (established in Lesson 1) and performs THREE actions: (1) ANSWER a previous sticky note question if today's evidence allows it — they write 'ANSWERED in L3' on the note and add a brief answer strip below it; (2) ADD at least one new question generated by today's findings (e.g., 'If the body can make some fats itself, why do we still need to eat fat?', 'Why does fat-soluble mean it needs fat to be absorbed?', 'Are there Kenyan foods that have BOTH protein AND fat?'); (3) REVISE their group's 'Chemical Map of Food' card, adding lipids and vitamins as new chemical categories with arrows showing their functions. Groups have 5 minutes for this. The teacher then facilitates a 3-minute whole-class DQB gallery walk — learners read new questions silently and place a dot sticker on the question they find MOST interesting or confusing. The top two dot-sticker questions are flagged as 'priority questions for Lesson 4.'	1. Direct groups to the DQB: 'Go to the board with your group. You have exactly 5 minutes for three jobs — answer what you can, add what you're wondering, update your chemical map card. Assign roles: one writer, one reader of old cards, one map updater.' 2. Circulate during the 5 minutes. Prompt groups that are only adding questions but not answering: 'Look at last lesson's questions about what makes foods different — can you now answer any of those with today's data?' [WAIT 10 seconds.] 3. Facilitate gallery walk: 'Now you have 3 minutes — silent walk, read the new questions, place your dot on the ONE question you find most interesting or most confusing.' [WAIT 3 minutes.] 4. Count dot stickers aloud with the class. Announce: 'These two questions — [read them aloud] — are our priority questions. We will specifically design Lesson 4 to answer them. Write them in your science notebook now.' 5. Connect back: 'Every question you add to this board is a gap in our understanding of WHY the Kisumu child's body is suffering. We are building the answer one lesson at a time.'	Strategy: Living Driving Question Board with Prioritisation Dots. The DQB is a dynamic public record of the class's evolving understanding. The dot-sticker prioritisation is a democratic, low-stakes formative tool that reveals collective confusion and curiosity without individual risk. Requiring groups to ANSWER old questions (not just add new ones) reinforces cumulative learning — learners see their own intellectual progress across lessons. Connecting new DQB questions explicitly to Lesson 4 creates anticipatory sets and maintains narrative momentum across the unit.	Observable indicators: (1) At least one old DQB question is marked 'ANSWERED in L3' per group — if no group can answer any previous question, the teacher should note this as a comprehension gap and plan a brief review at the start of Lesson 4. (2) New questions are conceptually specific (mention a chemical name, a body function, or a Kenyan food) rather than vague ('I want to know more about food') — prompt vague questions with: 'Can you name the chemical you're curious about?' (3) Chemical Map cards now include FOUR chemical categories (carbohydrates, proteins, lipids, vitamins) — cards missing lipids or vitamins indicate the group has not yet integrated today's learning into their conceptual model.
Model Building Phase 📹 **VIDEO:**	Groups retrieve their Cumulative Chemical Model of Food (a large A3 poster started in Lesson 1 and updated in Lesson 2). Today's	1. Distribute group models and say: 'Your model must now tell the story of ALL the chemicals we have found so far —	Strategy: Cumulative Model Revision with Peer Critique. Iterative model building is a core NGSS Science and Engineering	Observable indicators: (1) Models show the lipid structure (glycerol + fatty acids) — even a simple labelled diagram is sufficient;

Endomembrane system Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Meet the Family: Inv..:Segment 6 Single Letter Codes and DNA Codons for Amino Acids Source: MIT Blossoms 🔍 Search ARES for similar readings	task: REVISE AND EXPAND the model to include lipids and vitamins. Each group must add: (1) A 'Lipids' section showing the structure (glycerol backbone + 3 fatty acid chains, drawn simply), the function (energy storage, cell membrane component, fat-soluble vitamin carrier), and Kenyan food sources (groundnuts, avocado, coconut, milk); (2) A 'Vitamins' section organised into fat-soluble and water-soluble columns, with at least 2 examples each, their functions, and Kenyan food sources; (3) ARROWS connecting lipids to fat-soluble vitamins (showing that fat is needed to absorb Vitamins A, D, E, K); (4) A RED BORDER or 'MISSING FROM UGALI' label around the chemicals absent from the Kisumu child's ugali-only diet. Groups display their updated models at their stations. A 5-minute peer feedback round follows — each group visits one other group's model and leaves a sticky note with: one GREEN DOT comment ('I see evidence of ___') and one ORANGE DOT question ('I wonder why you ___'). Groups return and read their feedback before the lesson closes.	carbohydrates, proteins, lipids, vitamins. By the end of this unit it will be a complete chemical story of food. Add today's chemicals using the evidence from your data table and CER paragraph.' 2. Point to the specific model requirement: 'The ARROW connecting lipids to fat-soluble vitamins is non-negotiable — it is the most important scientific connection from today's lesson. Make it visible.' 3. Circulate and prompt: 'Show me on your model where the Kisumu child's diet is missing. Label it clearly — not just one missing chemical, but all the gaps you have found so far.' [WAIT 10 seconds per group.] 4. For the peer feedback round: 'You are not grading each other — you are helping each other see what is strong and what is missing. Green dot: specific evidence you can SEE in their model. Orange dot: a genuine question about their model.' 5. After groups read feedback, close with: 'Look at how much your model has grown since Lesson 1. You started with a question — why do two children eating the same amount of food end up with different bodies? Your model is becoming the answer. What chemical story does it tell so far?' Cold-call 2 individual learners for a 10-second response each. [WAIT 10 seconds per learner.] 6. Final celebration prompt: 'You didn't just learn ABOUT science today — you DID science. You tested real food, you built a real argument, and you revised a real scientific model. That is exactly what biochemists like those at the Kenya Medical Research Institute do every day.'	Practice (Developing and Using Models). By returning to the SAME poster across all 6 lessons, learners experience science as a process of progressive refinement rather than a set of isolated facts. The peer feedback structure (green dot / orange dot) introduces scientific peer review in an age-appropriate, non-threatening format. The 'MISSING FROM UGALI' red border keeps the anchoring phenomenon visually present in the model, preventing learners from treating the chemistry as abstract content disconnected from the real child in Kisumu.	absence of structure (only a function list) indicates surface-level understanding; prompt: 'What is lipid actually MADE OF at the chemical level?' (2) Arrow between lipids and fat-soluble vitamins is present and labelled — this is the lesson's key integrative concept; absence indicates the connection was not consolidated. (3) 'MISSING FROM UGALI' labels cover at minimum: proteins (from Lesson 2), lipids, and fat-soluble vitamins — models that only mark lipids suggest learners have not integrated Lesson 2 evidence into their current model. (4) Peer feedback sticky notes are specific rather than generic ('I see you drew fatty acids' vs. 'Good model') — collect sticky notes after class to gauge peer understanding quality.

D. TEACHER REFLECTION
After the lesson, reflect on the following questions in your professional journal: (1) CONTENT: Did learners successfully make the connection between dietary fat and fat-soluble vitamin absorption? This is the most conceptually demanding idea of the lesson — if fewer than 60% of groups included the connecting arrow in their models, plan a 5-minute re-teach at the start of Lesson 4 using a concrete analogy (e.g., 'Fat-soluble vitamins are like passengers — fat is the bus. No bus, no journey into the body'). (2) EQUITY: Were learners with nut allergies fully included in all practical activities? Did the avocado alternative provide equivalent data? (3) LANGUAGE: Did learners use precise chemical language in their CER paragraphs (naming specific vitamins, fatty acids, glycerol) or did they revert to everyday language ('fats and vitamins are good for you')? Plan vocabulary reinforcement if needed. (4) PHENOMENON CONNECTION: Did the 'MISSING FROM UGALI' model labels show that learners are tracking the phenomenon across lessons (listing protein gaps from Lesson 2 alongside today's lipid/vitamin gaps), or are learners treating each lesson as isolated content? If the latter, begin Lesson 4 with a 3-minute phenomenon recap — show the photographs of the two Kisumu children again and ask: 'What can our model now explain that it couldn't two lessons ago?' (5) SAFETY: Were all ethanol safety protocols followed? Were goggles worn throughout Activity 2? Note any safety incidents for future lesson planning and report as required by school safety policy.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners performed the translucent spot test and ethanol emulsion test on 8 Kenyan food samples (ugali, groundnuts, avocado, githeri, coconut flesh, whole milk, sukuma wiki, mandazi) and recorded which foods contain lipids. Learners observed that ugali tested negative for lipids in both tests, while groundnuts, avocado, coconut, and milk tested strongly positive. Learners also read and annotated a structured text on fat-soluble and water-soluble vitamins, identifying their roles and Kenyan food sources.
What did I learn?	Learners learned that lipids are composed of glycerol and fatty acids and serve as long-term energy stores, cell membrane components, and carriers for fat-soluble vitamins (A, D, E, K). They learned that vitamins are micronutrients needed in small quantities but critical for metabolic processes, immunity, and organ function. Critically, they learned that fat-soluble vitamins cannot be absorbed without dietary fat — meaning a fat-free diet causes both direct lipid deficiency AND indirect vitamin deficiency. Ugali's chemical profile now includes: rich in carbohydrates, low in protein (Lesson 2), and absent in lipids and fat-soluble vitamins (Lesson 3).
How does this explain the phenomenon?	Learners constructed CER (Claim–Evidence–Reasoning) paragraphs arguing that the Kisumu child eating only ugali is deficient in lipids and fat-soluble vitamins in addition to protein. They explained that this deficiency would cause multiple overlapping health effects — not only the swollen belly and thin limbs of kwashiorkor (protein deficiency, Lesson 2), but potentially also night blindness (Vitamin A), weak bones (Vitamin D), and impaired immunity. Learners updated their Cumulative Chemical Model of Food to include lipids and vitamins, connected by a structural arrow showing fat-soluble vitamin absorption dependency, and clearly labelled all chemical gaps in ugali.

LESSON 4 (80 minutes): Building Life's Molecules: Structure of Carbohydrates and Proteins

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To investigate the molecular structure of carbohydrates and proteins, and explain how differences in structure determine differences in function — connecting chemical structure directly to the visible health differences observed between the two Kisumu children.
Knowledge	Learners identify and describe the structural units of carbohydrates (monosaccharides, disaccharides, polysaccharides) and proteins (amino acids linked by peptide bonds), including the role of glucose → glycogen storage and amino acids → structural/functional proteins in the body.
Skills	Learners construct annotated molecular diagrams, build physical models of carbohydrate and protein molecules using locally available materials (bottle tops and straws), interpret guided reading passages, and map Kenyan food sources to each chemical class.
Attitudes	Learners appreciate that the invisible chemical composition of food — not just its quantity — determines health outcomes, developing empathy for communities like those in Kisumu affected by nutritional deficiencies.
Key Inquiry Question	What are the chemicals of life and what roles do they play in living organisms?
Purpose in Storyline	Lessons 1–3 established that food contains different chemical classes (carbohydrates, proteins, lipids, vitamins, minerals, water) and that food tests can detect their presence. Lesson 4 zooms into the molecular architecture of carbohydrates and proteins — the two classes most central to the anchoring phenomenon — so learners can explain WHY ugali (starch/polysaccharide only) cannot build or repair the body tissues that the Kisumu child with kwashiorkor is visibly losing. This lesson is the structural-understanding pivot of the unit: learners move from 'what is present' to 'how it is built and therefore what it can do.'
Safety Notes	No hazardous chemicals are used in this lesson. Bottle tops and straws are handled carefully — sharp edges on cut straws should be avoided. Learners with small siblings at home are reminded that bottle tops are a choking hazard and should be stored safely after use. Learners wash hands before and after any food-handling demonstrations.

B. LESSON OVERVIEW

Learners begin by revisiting the anchoring phenomenon photographs of the two Kisumu children and predicting — now with three lessons of prior evidence — what molecular difference between ugali and githeri might explain the health gap. They then engage in three layered evidence-gathering activities: (1) guided reading and annotated diagram construction on monosaccharides → disaccharides → polysaccharides; (2) physical model-building of a dipeptide using bottle tops (amino acids) and straws (peptide bonds); and (3) a Kenyan food-mapping activity that places familiar foods — ugali, githeri, nyama choma, milk, sukuma wiki — onto a molecular classification chart. Sensemaking occurs through a Gallery Walk of peer-built models and a whole-class discussion anchored to the driving question. The lesson closes with learners revising their personal Driving Question Board cards and their cumulative molecular models.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown the anchoring phenomenon photographs of the	Display the two photographs prominently. Say: 'Look carefully at	Think-Pair-Square-Share: This strategy activates prior knowledge	Teacher scans written predictions during circulation — look for

📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** DNA to RNA to Protein (PLIX) Source: CK-12 🔍 Search ARES for similar readings	two Kisumu children (the child with kwashiorkor eating ugali daily vs. the healthy child eating githeri) displayed at the front of the room. Working individually for 2 minutes, then in pairs for 2 minutes, learners write a prediction in their science notebooks responding to the prompt: 'We now know ugali is mostly starch and githeri contains both starch and protein. Predict: what do you think is DIFFERENT about the molecules in starch versus the molecules in protein — and why might that difference matter for the child's body?' Pairs then share predictions with a neighbouring pair (think-pair-square) before three pairs are cold-called to share with the class. Predictions are recorded on a class 'Before' column of a T-chart on the board titled 'What we predicted vs. what we found.'	these two children. You have spent three lessons gathering evidence about the chemicals in food. Today we go deeper — we look INSIDE those chemicals at their molecular structure. Before I show you anything new, I want your best scientific prediction.' Circulate during individual writing — do not correct or affirm. After 2 minutes, say: 'Now share with your partner. You have 2 minutes — both of you must speak.' After pair discussion, say: 'I am going to cold-call three pairs — be ready.' Cold-call Pair 1, Pair 2, Pair 3. For each response, ask a probing follow-up: 'What evidence from our previous lessons supports that prediction?' Allow 10–12 seconds of WAIT TIME after each follow-up before accepting a response. Record key prediction words (e.g. 'starch is simpler,' 'protein has more parts,' 'amino acids') verbatim on the T-chart 'Before' column without evaluation.	from Lessons 1–3, surfaces existing mental models about molecular structure, and creates cognitive dissonance that motivates the evidence-gathering phases. Recording predictions publicly commits learners to ideas they will test and revise, making the later sensemaking more meaningful.	evidence that learners are connecting chemical class names (carbohydrate, protein) to function rather than just food names. Observable indicator of readiness: learner writes at least one functional claim (e.g. 'starch gives energy but cannot build muscle'). Learners who write only food names (e.g. 'ugali has starch, githeri has beans') are noted for targeted questioning during Phase 2.
Observe Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** DNA to RNA to Protein (PLIX) Source: CK-12 🔍 Search ARES for similar readings	ACTIVITY A — Guided Reading & Annotated Diagrams (25 minutes): Learners receive a one-page illustrated reading passage (ARES platform resource) titled 'From Sugar to Starch: The Architecture of Carbohydrates' and a second passage titled 'Amino Acids: The Alphabet of Proteins.' Each passage includes structural diagrams (glucose ring, glycosidic bond forming maltose, glycogen branching diagram; generic amino acid structure, peptide bond formation, short polypeptide chain). Learners read individually for 8 minutes, then use coloured pencils to annotate their diagrams: (a) label monomer units in one colour, (b) label the bond type in a	Distribute reading passages and coloured pencils. Say: 'You have 8 minutes of silent reading. As you read, use your pencils — do not just highlight, ANNOTATE. I want to see your thinking on the page.' Set a visible timer. Circulate and look for annotation quality — if a learner is not annotating, crouch to their level and whisper: 'Show me with your pencil what a monomer is in that diagram.' After 8 minutes, say: 'Now annotate your diagrams using the three tasks on the board' (display tasks). Give 7 minutes. Transition to model building: 'In your groups, you are about to become molecular architects. Your bottle tops are amino acids. Your straws are peptide bonds. Build	Annotated Diagram + Physical Modelling (Dual Coding + Embodied Cognition): Annotating structural diagrams with colour-coded labels forces learners to distinguish monomers from polymers and bonds from molecules — a distinction that is frequently confused. Physical model-building with bottle tops and straws engages spatial and kinesthetic reasoning, making abstract molecular geometry concrete. Together, these strategies build a multi-representational understanding of molecular structure that learners can later deploy in explanation and model-building phases.	During annotation: teacher looks for correct colour-coding — specifically that glycosidic bonds and peptide bonds are labelled separately and not confused. Observable indicator: learner correctly labels a peptide bond between two amino acid bottle tops AND writes a function note distinguishing 'energy storage' (glycogen) from 'body building' (protein). During food mapping: observable indicator of understanding is correct placement of glycogen (polysaccharide, not protein) — this is a common error that reveals whether learners have read carefully. Flag any group that places glycogen under 'Protein' for

	second colour, (c) write a one-sentence function note next to each polymer. ACTIVITY B — Model Building (15 minutes): In groups of four, learners receive 20 bottle tops (representing amino acids — different colours for different R-groups), 10 straws cut to 3 cm lengths (representing peptide bonds), and a modelling instruction card. Groups construct: (i) a monosaccharide 'monomer' using a single bottle top with a label; (ii) a disaccharide by connecting two bottle tops with one straw; (iii) a polysaccharide chain of at least six bottle tops; (iv) a dipeptide and then a tripeptide using bottle tops of mixed colours. Groups photograph their models using the ARES tablet (if available) or sketch them into notebooks. ACTIVITY C — Kenyan Food Molecule Mapping (5 minutes): Each group receives a blank classification chart with columns: Monosaccharide / Disaccharide / Polysaccharide / Protein. They are given cards bearing the names: glucose (in sugarcane juice), sucrose (in chai), starch (in ugali), glycogen (stored in nyama choma meat), amino acids (in beans/githeri), keratin (in hair/nails), haemoglobin (in blood). Groups place each card in the correct column.	what I describe — step by step.' Read each build step aloud, demonstrate ONE step physically at the front using oversized props if available, then release groups to build. After 10 minutes of building, say: 'You have 2 minutes — sketch your final model in your notebook and write ONE sentence: what does this model tell us about why proteins are so varied?' Allow 12 seconds WAIT TIME after posing that question before groups write. For food mapping, distribute cards and say: 'Place each card in the correct column. Disagreements are good — argue with evidence from your reading.' After 4 minutes, cold-call two groups to justify one placement each.		targeted re-teaching during Phase 3.
Explain Phase 📹 **VIDEO:** [Dehydration synthesis or a condensation reaction] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos]	GALLERY WALK — Molecular Model Exhibition (10 minutes): Each group's bottle-top model is placed on their desk with a sticky-note label. Half the class circulates (Walkers) while the other half stands by their model as 'Explainers.' After 4 minutes, roles swap. Walkers must write one question and one connection on a	Initiate Gallery Walk: 'Explainers — your job is to explain your model to every Walker who visits. Use the words: monomer, polymer, bond, function. Walkers — leave a question AND a connection at every model.' Set 4-minute timer. After swap, set another 4-minute timer. Bring class back together: 'Let us build our explanation	Gallery Walk + Socratic Questioning Sequence: The Gallery Walk distributes the explanatory burden across all groups and forces learners to articulate molecular structure in their own words — a high-leverage retrieval and elaboration practice. The Socratic questioning sequence scaffolds learners from	Teacher reads 4–5 Sensemaking Summaries during the final minutes of the phase (circulate and photograph with tablet or read over shoulder). Observable indicator of proficiency: learner's summary (a) names at least one monomer for each macromolecule class, (b) identifies nitrogen as the key distinguishing element in

📖 **HTML:** [DNA to RNA to Protein (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	sticky note left at each model they visit. WHOLE-CLASS SENSEMAKING DISCUSSION (10 minutes): Teacher facilitates a structured discussion using the following sequence of questions, returning always to the anchoring phenomenon. Learners respond in full sentences, referencing their models and diagrams. The class builds a shared explanation on the board: 'Ugali provides starch (a polysaccharide of glucose). Glucose monomers release energy. BUT — glucose monomers contain only C, H, and O. Amino acids contain N (nitrogen) as well. The body cannot build proteins — including muscle, enzymes, and the protein albumin that prevents the swollen belly — without amino acids. Githeri provides amino acids from beans. Ugali alone does not. This is why the child eating only ugali develops kwashiorkor.' Learners write this explanation in their own words in their notebooks as a 'Sensemaking Summary.'	together. I am going to ask you three questions. Each answer gets us closer to solving the mystery of our two Kisumu children.' Q1: 'What elements are found in carbohydrates?' (Cold-call 1 learner — WAIT 10 seconds.) Q2: 'What extra element is found in amino acids that is NOT in glucose?' (Cold-call 1 learner — WAIT 12 seconds. Expected: nitrogen.) Q3 (to whole class, think-pair then share): 'If the Kisumu child eating only ugali has NO dietary source of amino acids — what happens to body tissues that need constant protein repair, like muscle and blood proteins?' (Pair discuss 90 seconds, then cold-call 2 pairs.) Scribe key words on board: C-H-O (carbohydrates) vs. C-H-O-N-S (proteins). Say: 'This is the molecular reason. Not quantity of food — CHEMICAL COMPOSITION of food. Write your own sensemaking summary now — 3–4 sentences, in your own words, connecting molecular structure to the child's condition.'	elemental composition → structural difference → physiological consequence → phenomenon explanation, mirroring the scientific reasoning chain. Writing the Sensemaking Summary consolidates understanding and produces a formative artefact.	amino acids, and (c) explicitly connects protein absence to a named symptom of the Kisumu child (swollen belly = loss of albumin protein, or thin limbs = loss of muscle protein). Learners who write only 'ugali has no protein' without molecular reasoning are identified for additional scaffolding in Phase 5.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Dehydration synthesis or a condensation reaction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [DNA to RNA to Protein (PLIX)] Source: CK-12	Learners return to the class Driving Question Board. Each learner receives two sticky notes of different colours — one colour for 'Questions this lesson ANSWERED' and one for 'New questions this lesson RAISED.' Individually (3 minutes), learners write: (a) one answered question connected to today's molecular structure learning (e.g. 'Why can't ugali alone build muscle? Because starch has no nitrogen-containing amino acids.') and (b) one new question raised (e.g. 'How does the body break polysaccharides back into glucose — and where does this happen?' or 'Are all	Say: 'Our Driving Question Board is a living record of our growing understanding. Today we go back to it. I want two sticky notes from each of you — one for what we can now ANSWER, one for what we still wonder.' Distribute sticky notes. Allow 3 minutes of silent individual writing. As learners post notes, circulate and read them — do not correct yet. Select three 'new question' notes that align with upcoming lessons (digestion, lipids, enzymes). Read them aloud: 'Someone asks: [read question]. Hands up — who else wonders this?' Count hands visibly. Say: 'These questions will	Driving Question Board Revision (Metacognitive Progress Tracking): Revisiting the DQB at the end of each lesson makes learning growth visible and tangible. Distinguishing 'answered' from 'new' questions trains learners to recognise the iterative nature of scientific inquiry — answering one question always generates new ones. Public voting on new questions builds collective ownership of the inquiry and previews upcoming lessons with genuine learner-generated curiosity.	Teacher photographs the DQB after sticky notes are placed. Observable indicators: (a) 'answered' notes reference molecular vocabulary from today (monomer, polymer, amino acid, peptide bond, glycosidic bond) — not just food names; (b) 'new question' notes are scientifically framed (begin with 'how,' 'why,' 'what happens when') rather than purely factual ('what is a protein'). Any learner whose 'answered' note is still at the food-name level (e.g. 'githeri has protein, ugali does not') is paired with a stronger peer for the model-building phase.

🔍 Search ARES for similar readings	proteins in githeri beans actually usable by the human body?'). Sticky notes are placed on the DQB. Teacher reads three 'new questions' aloud and the class votes (thumbs up) on which ones are most exciting — these become preview hooks for Lessons 5 and 6.	drive our next two lessons. Your curiosity is doing exactly what scientists' curiosity does — it points to the next investigation.' Point to the DQB and say: 'Notice how our board has grown since Lesson 1. Four lessons ago, many of you could not explain the difference between the two children. What can you explain now?'		
Model Building Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** DNA to RNA to Protein (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative unit model — a large annotated diagram they have been building since Lesson 1, representing 'What food is made of and what it does in the body.' Today, they add Layer 4: Molecular Structure. Using pencil and coloured pens, each learner adds to their personal model: (a) a carbohydrate section showing glucose → maltose → starch/glycogen with labelled glycosidic bonds and a note on function (energy release / energy storage); (b) a protein section showing 2–3 differently coloured amino acids connected by labelled peptide bonds, with a note on function (body-building, enzymes, transport proteins like haemoglobin); (c) a Kenyan food icon next to each chemical class (ear of maize → starch; beans/githeri → protein; sugarcane → glucose). In the corner of their model, learners draw a small 'Phenomenon Connection Arrow' pointing to a mini-sketch of the two Kisumu children and write one sentence: 'The child eating only ugali lacks [X] because ugali provides [Y] but not [Z].' Groups of four peer-review each other's models using a three-point checklist: (1) Are monomers and polymers both shown? (2) Is	Say: 'Your model is your most important scientific artefact in this unit. Today it grows again. Open to your model page. You are adding Layer 4 — Molecular Structure. I will give you 10 minutes. Use the board diagram and your annotated reading to guide you — do not copy, TRANSLATE the ideas into your own model.' Display a partially completed exemplar model on the board (teacher-drawn, deliberately imperfect — missing one bond label — so learners must improve on it). After 10 minutes, say: 'Now peer review in your group of four. Use the three-point checklist on the board. Be specific — do not just say 'good job.' Say: 'You labelled the glycosidic bond here but not here.' Give 5 minutes for peer review. Close by asking one learner to read their 'Phenomenon Connection' sentence aloud. Cold-call 2 learners. After the second reading, say: 'Four lessons ago, you could not explain those two children. Listen to what you just said. That is science — that is YOU doing science.'	Cumulative Model Revision (Progressive Knowledge Building): Requiring learners to add a new molecular layer to a model they have been building since Lesson 1 makes knowledge accumulation visible and prevents compartmentalisation. Each revision forces learners to integrate new structural knowledge with prior functional knowledge. The 'Phenomenon Connection Arrow' is a deliberate device to prevent the model from becoming an abstract exercise — every structural addition must be tethered back to the two Kisumu children, maintaining the explanatory purpose of the unit.	Collect 6 models (one per group, rotate which member's model is collected each lesson) for detailed written feedback. Observable indicators of proficiency in model: (a) glucose/amino acid monomers are visually distinct from their polymers (not just named but drawn as connected units); (b) glycosidic bond and peptide bond are both labelled and not confused with each other; (c) the Phenomenon Connection sentence names a specific protein function lost in the kwashiorkor child (e.g. albumin, muscle protein, or enzyme) — not just 'protein is missing.' Models that meet all three criteria are marked 'Proficient.' Models meeting only (a) are flagged for re-teaching in Lesson 5 warm-up.

	the bond type labelled? (3) Is there a Kenyan food example for each class? Learners initial each other's models to confirm peer review.			

D. TEACHER REFLECTION
After the lesson, reflect on the following questions in your professional journal: 1. PHENOMENON CONNECTION: Did learners' Sensemaking Summaries and Phenomenon Connection Arrows in their models explicitly name a molecular reason for the Kisumu child's symptoms — or did they remain at the food-name level ('ugali has no protein')? If most learners stayed at the food-name level, plan a 5-minute warm-up for Lesson 5 using cold-call questioning that pushes from food names to molecular vocabulary. 2. MODEL BUILDING: Were learners able to distinguish glycosidic bonds (carbohydrates) from peptide bonds (proteins) in their diagrams and bottle-top models? Confusion between these two bond types is a strong diagnostic signal that the reading passages and model-building activity need additional scaffolding — consider providing a bond-type reference card at the start of Lesson 5. 3. DQB QUALITY: Did the 'new questions' on the Driving Question Board advance toward the next lessons (digestion, enzyme action, lipid structure) — or did they circle back to already-answered questions? If learners' questions are not advancing, the Gallery Walk may not be generating enough cross-pollination of ideas — consider restructuring Explainer roles in the next Gallery Walk so that groups must present to a partner group, not just to Walkers passing by. 4. EQUITY CHECK: Which learners were not cold-called today? Ensure your cold-call record is balanced across gender, home language, and prior performance. The molecular structure content in this lesson (bond types, elemental composition) tends to advantage learners with stronger literacy — were hands-on model builders who struggle with reading given sufficient opportunity to demonstrate understanding through their bottle-top models? 5. KENYAN CONTEXT: Did learners successfully map githeri, ugali, nyama choma, and sukuma wiki onto the molecular classification chart — or did they struggle to move from food name to chemical class? If the food-mapping activity felt disconnected, consider opening Lesson 5 with a brief cooking analogy: 'When your mother makes githeri, she is combining a polysaccharide (maize) with a protein source (beans) in the same pot — your digestive system does something similar at the molecular level.'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners constructed annotated diagrams of carbohydrate and protein molecular structures from guided reading passages, built physical models of monosaccharides, disaccharides, polysaccharides, and dipeptides using bottle tops and straws, and mapped Kenyan foods (ugali, githeri, nyama choma, sugarcane juice, chai) to their correct molecular classification (monosaccharide, disaccharide, polysaccharide, protein).
What did I learn?	Carbohydrates are made of monosaccharide monomers (e.g. glucose) joined by glycosidic bonds to form disaccharides (e.g. sucrose in chai) and polysaccharides (e.g. starch in ugali, glycogen stored in muscle/liver). Proteins are made of amino acid monomers joined by peptide bonds; amino acids contain nitrogen (N) in addition to C, H, and O — an element absent in carbohydrates. This elemental and structural difference means starch can supply energy but cannot provide the building blocks for muscle, blood proteins (haemoglobin), or albumin. A diet of ugali alone provides no amino acids, so the body cannot build or repair proteins — explaining the swollen belly (loss of albumin), thin limbs (loss of muscle protein), and discoloured hair (loss of keratin) seen in the Kisumu child with kwashiorkor.
How does this explain the phenomenon?	The two Kisumu children eating similar quantities of food have completely different bodies because the MOLECULAR COMPOSITION of their food differs, not just the amount. Ugali is almost entirely starch — a polysaccharide composed only of glucose monomers containing C, H, and O. Githeri adds beans, which are rich in protein — polymers of amino acids containing N. Without amino acids, the child's body cannot synthesise the proteins essential for tissue building, immune function, and fluid

	balance. The anchoring phenomenon (kwashiorkor) is therefore a direct consequence of molecular absence: no amino acids in the diet → no protein synthesis → visible, devastating loss of body structure and function. Food quantity is irrelevant if the molecular composition is incomplete.

LESSON 5 (40 minutes): Sensemaking: Connecting the Chemistry of Food to Health Outcomes in Kisumu	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners use evidence gathered from food tests and molecular structure investigations to construct and communicate scientific explanations for why two children eating the same quantity of food have completely different health outcomes, anchoring all reasoning to the chemicals of life.
Knowledge	Learners will be able to: (1) explain that ugali provides carbohydrates but lacks sufficient protein, leading to kwashiorkor in the Kisumu child; (2) match each deficiency disease (kwashiorkor, marasmus, scurvy, rickets, anaemia) to the specific missing chemical of life; (3) explain how albumin loss causes oedema (swollen belly), keratin loss causes hair discolouration, and muscle protein loss causes thin limbs.
Skills	Learners will: analyse and interpret food test data from Lesson 4 to construct evidence-based explanations; evaluate dietary composition using chemical evidence; communicate scientific reasoning orally and in writing; update the Driving Question Board with evidence-supported answers.
Attitudes	Learners will demonstrate curiosity about the chemical basis of health, empathy for communities affected by malnutrition in Kenya, and confidence in using scientific evidence to explain real-world phenomena.
Key Inquiry Question	What are the chemicals of life and what roles do they play in living organisms?
Purpose in Storyline	This is the Explain lesson — the sensemaking pivot of the entire unit. Learners have now gathered sufficient evidence (molecular structures in Lessons 2 and 3, food test data in Lesson 4) to construct a full scientific explanation for the anchoring phenomenon. This lesson transforms accumulated observations into causal understanding: the child with kwashiorkor is not food-insecure in quantity but is chemically deficient in protein, and the githeri child thrives because maize and beans together supply both carbohydrates and essential amino acids. All six chemicals of life are connected to health consequences, preparing learners for the model-building consolidation in Lesson 6.
Safety Notes	No laboratory chemicals are used in this lesson. If photographs of children with kwashiorkor or other deficiency diseases are displayed, the teacher should frame them with sensitivity and respect, reminding learners that these conditions are preventable and that real families are affected. Avoid stigmatising language.

B. LESSON OVERVIEW

Learners open the lesson by revisiting the photographs of the two Kisumu children and their Lesson 4 food test results table. In the Predict phase, they are asked what they now think is the single most important chemical difference between ugali and githeri. In the Observe phase, groups analyse their food test data alongside a provided deficiency disease matching card. In the Explain phase, each group constructs a written scientific explanation for the anchoring phenomenon, presents it to the class, and the teacher facilitates whole-class sensemaking discussion linking findings to Kenyan public health nutrition campaigns (e.g. the Kenya National School Feeding Programme). The DQB is updated with answered questions. The lesson closes with a brief model-building preview task in which learners identify one gap remaining in their understanding before Lesson 6.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Bacterial Diseases - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown the two photographs of the Kisumu children displayed at the front of the room — the same images used in Lesson 1. The teacher returns their Lesson 4 food test results tables (or learners retrieve their notebooks). Working in their established groups of four, learners spend 3 minutes silently reviewing their food test results and then answer one prompt written on the board: 'Based on your food test results, what is the single most important chemical difference between ugali and githeri — and how does that difference explain what you see in the two children?' Each learner writes an individual prediction sentence before group discussion begins. Groups share their prediction sentences verbally within the group.	The teacher displays the two photographs and says: 'Look carefully at these two children from Kisumu. You saw them in Lesson 1 — but now you have evidence you did not have then. You have your food test data. You know what carbohydrates, proteins and lipids look like at the molecular level. So I want you to think harder than you did in Lesson 1.' The teacher writes the prompt on the board and says: 'Read the prompt. Write your own sentence first — do not talk yet. You have 2 minutes of silent thinking time.' WAIT TIME: 2 minutes of enforced silence while learners write. After 2 minutes the teacher says: 'Now share your sentence with your group. You have 1 minute.' After sharing the teacher asks 2 to 3 groups to read their prediction sentences aloud and records key words (protein, amino acids, nitrogen, kwashiorkor) on the board without evaluating them yet, saying: 'I am not telling you if you are right yet — we are going to use your evidence to find out.'	Activating prior knowledge through structured prediction — learners are required to connect two evidence sources (visual phenomenon plus food test data) before instruction, surfacing the conceptual link between protein absence and observed symptoms.	Teacher circulates during silent writing and reads learner sentences over shoulders to identify whether learners are connecting protein absence (not just food quantity) to the health difference. Learners who write only 'ugali has more starch' without mentioning protein are flagged for targeted questioning during the Observe phase.
Observe Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Bacterial Diseases -	Each group receives a Deficiency Disease Matching Card — a structured card with five rows (kwashiorkor, marasmus, scurvy, rickets, anaemia) and four columns (disease name, missing chemical of life, visible symptoms, Kenyan food that could prevent it). Two columns are pre-filled (disease name and visible symptoms) and two are blank (missing chemical and Kenyan food source). Learners use their Lesson 2 and 3 notes and their Lesson 4 food test data table to	The teacher distributes the Deficiency Disease Matching Cards and says: 'Your food test data is your evidence. Your molecular structure notes are your background knowledge. Use both to fill in the two blank columns on this card. Start with kwashiorkor because that is the disease we can see in our Kisumu child.' WAIT TIME: Allow 6 minutes for group work without interruption. The teacher circulates, listening to group talk. If a group is stuck on scurvy, the teacher prompts:	Evidence-to-explanation scaffolding — the partially completed Deficiency Disease Matching Card structures the cognitive task so learners focus on the causal link (missing chemical causes specific symptoms) rather than rote recall. The diet audit task operationalises the phenomenon as a real dietary analysis.	The teacher checks that learners correctly identify protein (specifically essential amino acids) as the missing chemical in kwashiorkor, and that their Biuret test evidence is cited in their reasoning. Misconception to watch for: learners attributing kwashiorkor to 'not enough food' rather than 'not enough protein specifically.' If this appears, the teacher re-asks: 'The child with kwashiorkor is eating the same amount of food in quantity as the healthy child. So what is different?'

Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	complete the blank columns collaboratively. For kwashiorkor specifically, they must cite their Biuret test result on the beans extract (purple colour = protein present) and connect the absence of such a result for ugali to the protein-deficient diet. Groups also calculate a simple diet audit: they are given a typical daily menu for each Kisumu child (child 1: ugali three times, child 2: githeri twice and ugali once) and must list which chemicals of life are present and which are absent or insufficient in each menu.	'Which chemical of life did we test with DCPIP in Lesson 4? What did you find in orange juice?' If a group conflates marasmus and kwashiorkor, the teacher asks: 'What is the difference between getting no food at all and getting food that is only starch? Which child in our phenomenon is closer to which disease?' After 6 minutes the teacher pauses the class and cold-calls one group per disease row to share their answer, asking the class: 'Does your card agree or disagree? Correct your card now if needed.' The teacher writes the completed table on the board as a class reference.		
Explain Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Bacterial Diseases - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group constructs a written scientific explanation (5 to 7 sentences) for the anchoring phenomenon using a provided sentence-starter framework: (1) 'Ugali is made mainly of... because our iodine test showed...'; (2) 'Githeri contains both... and... because our Biuret test showed...'; (3) 'The child eating only ugali develops kwashiorkor because...'; (4) 'Proteins are needed in the body to build... including albumin (which prevents oedema), keratin (hair), and muscle fibres...'; (5) 'Therefore the two children have different bodies because...'. One group spokesperson reads the completed explanation aloud. The teacher facilitates a whole-class discussion in which learners evaluate each other's explanations. The teacher then connects the class findings to the Kenya National School Feeding Programme and to the work of nutritional scientists at the Kenya Medical Research Institute (KEMRI) who study childhood	The teacher says: 'You now have enough evidence to answer the driving question — at least part of it. Your group is going to write a scientific explanation. I am not asking you to guess. I am asking you to use your food test results, your molecular structure knowledge, and your deficiency disease card as evidence. Use the sentence starters on the board.' WAIT TIME: Allow 5 minutes for group writing. After groups present, the teacher says: 'Listen carefully to each group. When they finish, I want you to say one thing you agree with and one thing you would add or change.' After all groups have presented, the teacher synthesises: 'Every group arrived at the same core idea: the Kisumu child with kwashiorkor is not starving in terms of energy — ugali gives energy through carbohydrates — but is starving in terms of the building-block molecules that only proteins can supply. That is a profound scientific insight and it is exactly	Collaborative explanation construction using sentence-starter scaffolds ensures all learners produce evidence-based reasoning rather than opinion. Peer evaluation of group explanations deepens critical thinking. Connecting to KEMRI scientists and the Kenya National School Feeding Programme grounds the science in authentic Kenyan societal context.	Teacher collects the written group explanations at the end of the lesson to assess whether learners: (1) correctly cite food test evidence (iodine for starch, Biuret for protein); (2) identify specific proteins (albumin, keratin, muscle protein) and explain their functions; (3) make a clear causal link between protein absence and kwashiorkor symptoms. A three-point rubric is used: 1 = identifies protein as missing; 2 = connects protein absence to specific symptoms; 3 = cites food test evidence and names specific proteins and their functions.

	malnutrition in Nyanza and Western Kenya. Learners update their individual DQB sticky notes: answered questions are moved to an 'Answered' column and any new questions are added.	what nutritional scientists at KEMRI and the World Food Programme have been telling Kenyan communities for decades.' The teacher then asks: 'What does this tell us about what food is really made of?' — re-echoing the driving question — and allows learners to respond freely for 2 minutes before summarising.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Bacterial Diseases - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their individual DQB sticky notes from the board (posted in Lesson 1). Each learner reads their original Lesson 1 question and decides: (a) Can I now answer this question using evidence from Lessons 1 to 5? If yes, the learner writes a brief answer on the back of the sticky note and moves it to the 'Answered' column on the DQB. (b) If the question is still partially unanswered or has generated a new question, the learner writes a new sticky note for the 'Still Wondering' column. The class briefly reviews which questions have been answered (typically questions about what food is made of, what causes kwashiorkor, and what proteins do) and which remain (typically questions about DNA, vitamins in detail, or how the body digests food). The teacher acknowledges remaining questions as the focus for Lesson 6 model building.	The teacher says: 'Go to the DQB. Find your sticky note from Lesson 1. Read your question. Ask yourself honestly: can I answer this now using evidence? If yes, write your answer on the back and move it to Answered. If not, write a new question on a fresh sticky note and add it to Still Wondering.' WAIT TIME: Allow 3 minutes for learners to work independently at the board. After the 3 minutes the teacher reads out 3 to 4 answered questions and says: 'Look how far we have come since Lesson 1. You could not answer these questions then. Now you can — and you can back your answers with chemical evidence.' The teacher then points to the Still Wondering column and says: 'These remaining questions are exactly what we will address in Lesson 6 when we build our final model. So do not worry — we are not done yet.'	The DQB update provides a metacognitive checkpoint — learners explicitly compare their current understanding to their Lesson 1 baseline, making conceptual growth visible and celebrated. This sustains motivation and models how science progresses through evidence accumulation.	The teacher scans the Answered column to check that learners are providing evidence-based answers (not just 'yes I know this now') and reads a sample of the Still Wondering questions to identify persistent misconceptions or genuine curiosity to address in Lesson 6.
Model Building Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan	As a closing preview task for Lesson 6, each group receives a blank A3 template titled 'Our Chemicals of Life Model' with six labelled boxes (one per chemical of life). Groups spend the final 4 minutes filling in whatever they already know for each box: structure, function, Kenyan food	The teacher distributes the A3 template and says: 'This is the beginning of your final model — the one you will complete and present in Lesson 6. Fill in whatever you are confident about right now. Put a star on anything you are not sure about. You are not expected to finish this today —	The partial model-building task serves as a forward-looking formative assessment and motivational bridge to Lesson 6. By making knowledge gaps visible through stars, learners develop self-regulated learning awareness. Returning to the photographs at closure re-anchors all chemistry to	Teacher notes which chemical-of-life boxes are consistently left blank or starred across groups and uses this to plan the Lesson 6 consolidation emphasis. If nucleic acids are universally blank, the teacher plans a brief targeted review at the start of Lesson 6 before model building begins.

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Bacterial Diseases - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	source, and consequence of deficiency. This is intentionally incomplete — it will be finalised in Lesson 6. Groups are asked to put a star next to any box that still feels uncertain. This gives learners a concrete starting point for the consolidation lesson and surfaces remaining knowledge gaps.	that is what Lesson 6 is for.' WAIT TIME: Allow 4 minutes of group work. The teacher circulates and notices which boxes are left blank or starred most frequently (typically nucleic acids, vitamins, and minerals) and says: 'I can see that most groups have stars on nucleic acids and vitamins — perfect. That tells us exactly what we need to focus on in our final lesson.' The teacher closes the lesson by returning to the two photographs and saying: 'Next lesson, your model will be able to explain everything we see in these two children — and everything we do not see, at the molecular level inside their cells. That is the power of understanding the chemicals of life.'	the human phenomenon.	

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Were learners able to move beyond 'ugali has less food' to the precise chemical explanation — protein absence leading to specific molecular consequences (albumin, keratin, muscle protein loss)? If not, what additional scaffolding is needed in Lesson 6? (2) Did the food test evidence from Lesson 4 genuinely drive the explanations, or did learners rely on memorised facts without citing evidence? If the latter, build an explicit evidence-citation practice into Lesson 6 model sharing. (3) Were there learners who showed empathy fatigue or distress when discussing the Kisumu children? Ensure Lesson 6 frames the phenomenon constructively — these conditions are preventable with nutritional knowledge. (4) Which DQB questions in the Still Wondering column are most common? Use these to sequence the Lesson 6 model-building discussion. (5) Did learners connect the science to their own diets or family practices? If so, this is a high-value transfer moment to honour in Lesson 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In Lesson 4, learners observed that iodine turned blue-black in ugali (confirming starch) but the Biuret test on ugali extract showed no purple colour (no protein), while the Biuret test on beans extract turned purple (protein present). Food test data showed githeri contains both starch and protein, while ugali contains starch only.
What did I learn?	Learners learned that kwashiorkor is caused specifically by protein deficiency — not by insufficient food quantity. The child eating ugali alone receives carbohydrates for energy but no amino acids, so the body cannot synthesise albumin (causing oedema/swollen belly), keratin (causing hair discolouration), or muscle proteins (causing thin limbs). Githeri provides essential amino acids from beans that complement the maize carbohydrate, explaining the healthy child's better health outcome. Five deficiency diseases were matched to their missing chemical of life: kwashiorkor (protein), marasmus (total caloric energy including protein and carbohydrate), scurvy (vitamin C), rickets (vitamin D and calcium), anaemia (iron).

How does this explain the phenomenon?	Learners constructed evidence-based scientific explanations linking their food test results and molecular structure knowledge to the anchoring phenomenon. The explanation demonstrates that food is not uniform — it is a mixture of specific chemicals of life, each with irreplaceable structural and functional roles in the body. The quantity of food matters less than its chemical composition, which is why two children eating the same amount of food can have completely different bodies and health outcomes.

LESSON 6 (40 minutes): Building Our Chemicals of Life Model: Consolidating the Science Behind Two Children in Kisumu	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners synthesise all prior learning from Lessons 1 to 5 into a single coherent visual model that explains why two children in Kisumu eating the same quantity of food can have completely different bodies and health outcomes, connecting molecular chemistry to visible biological consequences.
Knowledge	Learners will be able to: (1) identify the six major chemicals of life and state the molecular structure, biological function, Kenyan food sources and deficiency consequence of each; (2) explain why kwashiorkor results specifically from protein deficiency despite adequate carbohydrate intake; (3) distinguish between the nutritional profiles of ugali and githeri at the chemical level; (4) match five deficiency diseases to their missing chemical of life.
Skills	Learners will demonstrate the ability to: construct and annotate a cumulative visual scientific model integrating structure, function, food sources and deficiency data; evaluate a Kenyan daily menu for nutritional completeness; communicate scientific explanations using evidence from food tests, molecular diagrams and deficiency disease data; compare Lesson 1 predictions with final evidence-based understanding.
Attitudes	Learners will appreciate the personal and social relevance of biochemistry to Kenyan community health, show intellectual honesty in acknowledging how their initial predictions were incomplete, and demonstrate collaborative responsibility when co-constructing and presenting group models.
Key Inquiry Question	What are the chemicals of life and what roles do they play in living organisms, and how does understanding these chemicals explain the difference in health between two children in Kisumu eating the same amount of food?
Purpose in Storyline	This is the final consolidation lesson. Learners move from explanation (Lesson 5) to model building and synthesis. The anchoring phenomenon is fully resolved: learners can now state, with molecular evidence and food test data, exactly why the ugali child developed kwashiorkor while the githeri child remained healthy. The DQB is closed by celebrating answered questions and identifying any remaining curiosity questions for future learning.
Safety Notes	No hazardous chemicals are used in this lesson. If learners bring food samples for the menu evaluation task, remind them not to consume any samples used in class activities. Ensure all printed photographs of children with kwashiorkor are handled sensitively and discussed with respect for dignity.

B. LESSON OVERVIEW

In this 40-minute final lesson learners work in groups to complete and present a cumulative visual model — a poster or annotated diagram — showing the six chemicals of life, their structures, functions, Kenyan food sources and deficiency consequences, all anchored to the two Kisumu children from the opening phenomenon. Groups compare their Lesson 1 predictions with their current evidence-based understanding and identify what changed in their thinking. The class then reviews the Driving Question Board together, celebrating answered questions and noting any new questions generated. The lesson closes with an individual formative assessment task in which each learner evaluates a typical Kenyan daily menu, identifying which chemicals of life are present, missing or insufficient and predicting the health consequences.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Meet the Family: Inv..:Segment 6 Single Letter Codes and DNA Codons for Amino Acids Source: MIT Blossoms 🔍 Search ARES for similar readings	Learners spend the first 5 minutes revisiting their Lesson 1 predictions written on sticky notes on the Driving Question Board. Each group receives their original Lesson 1 prediction card — a brief written response to the question: What do you think food is really made of, and why does the same amount of different foods cause different health outcomes? Learners read their original prediction silently, then discuss in their groups: What did we believe then? What do we know now? What was wrong or incomplete in our first ideas? Groups identify two specific pieces of evidence from Lessons 2 to 5 that changed their thinking. A recorder in each group writes a one-sentence Before and After statement on a small card, for example: Before, we thought food was just energy. Now we know food is made of six specific chemicals each with a different molecular structure and function, and missing even one causes a specific disease.	Display the original anchoring photographs of the two Kisumu children at the front of the room. Say: We began this unit with a puzzle — two children, same village, same amount of food, completely different bodies. Before we build our final models today, I want you to go back to where you started. Read your Lesson 1 prediction card quietly. [WAIT TIME: 60 seconds of silent reading.] Now tell your group — what did you believe then, and what do you know now? [WAIT TIME: 2 minutes of group discussion.] Ask two or three groups to share their Before and After statements. Affirm the intellectual growth without dismissing original ideas. Say: Science always begins with incomplete ideas — what matters is that evidence changes our thinking. That is exactly what you have done across this unit.	Activating prior knowledge and metacognitive reflection — learners explicitly compare their naive conceptions with their current evidence-based understanding, making conceptual change visible and deliberate.	Teacher listens to group discussions and reads Before and After cards. Look for evidence that learners have moved from vague ideas about food as energy or nutrients to specific language about molecular structures, chemical classes and deficiency consequences. Identify any persistent misconceptions to address during the Explain phase.
Observe Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Meet the Family: Inv..:Segment 6 Single Letter	Groups spend 10 minutes completing their cumulative visual model using a pre-structured A3 poster template with six columns — one per chemical of life — and four rows labelled: Molecular Structure (draw or describe), Biological Function in the body, Kenyan Food Sources, and Consequence of Deficiency (name the disease and describe the visible symptom). Learners should also include a central anchor image of the two Kisumu children with arrows linking the ugali child to carbohydrate excess and	Distribute the pre-structured A3 poster templates and remind groups of the four rows they must complete for each chemical. Say: Your model must tell the story of the two children in Kisumu. Anyone reading your poster should be able to understand why one child is sick and one is healthy, just by looking at your model. Circulate and use targeted prompts: For carbohydrates, ask: Where exactly does glucose come from in ugali, and where does it go in the body? [WAIT TIME: 20 seconds.] For proteins, ask: Which	Model construction as sensemaking — learners do not simply recall information but must organise it spatially and causally, making explicit the links between molecular structure, biological function and health outcome. The central anchor image of the two children ensures the phenomenon remains the organising principle of the model.	Teacher checks each group poster for: (1) all six chemicals present; (2) molecular structure described or drawn for at least carbohydrates, proteins and lipids; (3) deficiency column completed with disease name and visible symptom; (4) the two Kisumu children referenced in the model. Note which groups are omitting nucleic acids, vitamins or minerals and prompt accordingly.

Codes and DNA Codons for Amino Acids Source: MIT Blossoms 🔍 Search ARES for similar readings	protein deficiency and the githeri child to balanced intake. Groups use their notes, comparison tables from Lesson 3 and food test results from Lesson 4 as reference materials. The teacher circulates and prompts groups to ensure all six chemicals are represented and that the deficiency column is completed with both the disease name and a specific visible symptom. Learners who finish early are asked to add a seventh column titled Water showing its role as a solvent and transport medium in cells.	amino acids does the body need and why can it not make them itself? [WAIT TIME: 20 seconds.] For lipids, ask: Can you remember the ethanol emulsion result for avocado — what did that tell us? [WAIT TIME: 20 seconds.] For the deficiency column, ask: What specific protein cannot be made without amino acids that causes the swollen belly? [WAIT TIME: 30 seconds — expecting learners to recall albumin.] If a group leaves nucleic acids vague, prompt: What would happen to cell division if there were no DNA to carry the genetic code?		
Explain Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Meet the Family: Inv..:Segment 6 Single Letter Codes and DNA Codons for Amino Acids Source: MIT Blossoms 🔍 Search ARES for similar readings	Each group has 2 minutes to present their completed model to the class, focusing on one key explanation: Why does the child eating only ugali develop kwashiorkor while the child eating githeri remains healthy — and what does this tell us about the chemicals of life? Presenters must use the language of molecular structure and biological function, not just food names. After all groups present, the teacher leads a 5-minute whole-class synthesis discussion. The class co-constructs a single master answer to the driving question on the board, drawing one contribution from each group. Learners copy the master answer into their science notebooks as a unit summary statement. The teacher also connects the learning to Kenyan public health by asking learners to think about how national school feeding programmes and nutrition campaigns — such as those promoted by the Kenya Nutritionists and Dietitians Institute	Before presentations begin, say: When your group presents, I want to hear the words molecular structure, amino acids, protein synthesis and deficiency. These are the scientific ideas that explain the phenomenon — not just the foods. [WAIT TIME: 30 seconds for groups to review their posters before presenting.] After all presentations, ask the class: Has every group agreed on why the ugali child is sick? [Pause for responses.] Now say: I want us to write one master answer together. What is the single most important scientific idea that solves our driving question? [WAIT TIME: 45 seconds of think time before taking contributions.] Write learner contributions on the board and shape them into a coherent explanation: Ugali provides only carbohydrates in the form of starch, which the body breaks into glucose for energy. Without proteins from food, the body cannot synthesise the amino acids needed to build albumin, keratin or muscle proteins, resulting in	Gallery-style group presentations followed by collective sensemaking — the class synthesises across group models to arrive at a shared, evidence-based scientific explanation. The master answer written collaboratively on the board makes the collective knowledge product visible and durable.	During presentations, listen for correct use of: albumin (oedema), keratin (hair discolouration), muscle proteins (thin limbs), essential amino acids (cannot be synthesised by the body), peptide bonds (protein structure). Use a quick show-of-hands confidence check after the master answer is written: Thumbs up if you can explain the phenomenon using molecular language, thumbs sideways if you are still unsure about one chemical, thumbs down if you need more support. Use thumb-sideways and thumb-down responses to identify learners for targeted follow-up.

	— use exactly this chemical understanding to design balanced meals for children across Kenya.	kwashiorkor. Githeri combines maize carbohydrates with bean proteins, providing essential amino acids and preventing deficiency. Say: That is biochemistry solving a real health problem in our community.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Meet the Family: Inv..:Segment 6 Single Letter Codes and DNA Codons for Amino Acids Source: MIT Blossoms 🔍 Search ARES for similar readings	The class gathers around the Driving Question Board as a whole group. The teacher reads each question on the board aloud — questions added by learners across Lessons 1 to 5 — and the class decides together: Answered, Partially Answered, or New Question Generated. Answered questions are moved to a Celebrated Answers section and marked with a star sticker. Partially answered questions are noted for future investigation. Any new questions generated by the model-building process — for example, How do cells use DNA to know which proteins to build? or Why do some vitamins dissolve in fat but not water? — are added to a New Questions column. Learners celebrate the intellectual journey by counting how many questions they answered as a class. The DQB moment is brief (4 minutes) but emotionally significant — it marks the closure of the unit and validates learner-generated inquiry.	Stand beside the DQB and say: Let us look at the questions you have been asking since Lesson 1. Some of these were written when you did not yet know what food was made of. Let us see how far we have come. Read each question clearly and pause for learners to respond. When a question is confirmed as answered, say: Who can give the answer in one sentence using scientific language? [WAIT TIME: 20 seconds.] Celebrate each answer enthusiastically. For questions that generated new curiosity, say: This is what real scientists do — every answer opens new questions. These are yours to carry forward. If the question How do cells know which protein to build? arises, say: That question connects to genetics and gene expression — you will investigate that in a future unit. It is a brilliant question to be holding.	DQB review as metacognitive closure — learners experience the full arc of inquiry from initial puzzlement to evidence-based resolution. Celebrating answered questions reinforces that scientific inquiry is productive and that learner questions drive real learning.	The teacher assesses the quality of one-sentence answers given by learners for each celebrated question. Look for: use of chemical names rather than food names alone; causal language (because, therefore, which means); reference to the two Kisumu children as the phenomenon that motivated the question. Note any questions that the class cannot answer fluently — these indicate gaps requiring teacher follow-up in the next sub-strand.
Model Building Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum)	In the final 10 minutes learners complete an individual formative assessment task called the Kisumu Menu Evaluation. Each learner receives a card showing a typical Kenyan daily menu: Breakfast — uji (thin maize porridge) with sugar; Lunch — ugali with sukuma wiki; Supper — rice with a small piece of fried fish.	Distribute the Kisumu Menu Evaluation cards and say: This is your individual task — no group discussion for this one. Use everything you have learned and your model to evaluate this menu. Think like a nutritionist advising a family in Kisumu. [WAIT TIME: allow 8 full minutes of individual working time with no interruption.]	Transfer task as culminating sensemaking — learners apply their integrated understanding to a novel but locally grounded scenario, demonstrating whether they can transfer chemical knowledge from the classroom phenomenon to a real-world nutritional evaluation task. The final prompt connecting the menu	Evaluate each learner response against four criteria: (1) Correct identification of at least four chemicals present in the menu with food evidence; (2) Correct identification of at least one nutritional gap with chemical specificity, for example insufficient protein because fish is a very small portion and no legumes are

🔍 Search ARES for similar videos 📖 **PDF:** Meet the Family: Inv..:Segment 6 Single Letter Codes and DNA Codons for Amino Acids Source: MIT Blossoms 🔍 Search ARES for similar readings	Learners must: (1) identify which chemicals of life are present, using the food test knowledge from Lesson 4 as evidence; (2) identify which chemicals are missing or insufficient; (3) predict one health consequence of any nutritional gap they identify; (4) suggest one affordable, locally available food that could be added to improve the menu, with a justification based on its chemical composition. Learners write their responses individually in their notebooks. The task is collected or photographed by the teacher at the end of the lesson for formative review.	Circulate silently, noting whether learners are applying chemical names and food test reasoning or reverting to general healthy eating language. At the 8-minute mark, say: In the last 2 minutes, write one sentence connecting your menu evaluation back to our driving question — what does this menu tell us about what food is really made of? Collect notebooks or take a photograph of each learner response before they leave. Close the lesson by saying: You came into this unit asking why two children eating the same amount of food could have such different bodies. You can now answer that question with the language of biochemistry, with evidence from food tests, with molecular diagrams and with real Kenyan food. That is the power of science.	to the driving question ensures the phenomenon anchors even the assessment moment.	included; (3) A plausible and chemically justified health consequence, for example risk of protein deficiency if the fish portion is consistently small; (4) A specific, affordable Kenyan food recommendation with a chemical justification, for example adding beans to lunch would provide essential amino acids needed for protein synthesis. Learners who score 4 out of 4 are ready for Sub-strand 1.3. Learners scoring 2 or below should receive a targeted revision card before the next lesson.

D. TEACHER REFLECTION

After the lesson, consider the following questions: Did all learners reach the level of molecular language when explaining the anchoring phenomenon, or did some revert to general food group language such as proteins are important for growth without explaining why at the amino acid and peptide bond level? Were there any chemicals of life — particularly nucleic acids or water — that groups consistently left thin or underdeveloped in their models? How did the Kisumu Menu Evaluation responses reveal the quality of transfer — could learners move from the specific ugali-versus-githeri scenario to a new food context? Were there questions on the DQB that the class could not answer fluently, and do these indicate gaps in the instructional sequence that need to be revisited? Finally, consider whether the lesson successfully honoured the dignity of the children depicted in the anchoring phenomenon — nutritional deficiency photography must always be handled with cultural sensitivity and care.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed their own Lesson 1 prediction cards alongside the completed group visual models showing all six chemicals of life with their structures, food sources, functions and deficiency consequences. Learners also observed the Kisumu Menu Evaluation card representing a realistic Kenyan daily diet.
What did I learn?	Learners learned that the six chemicals of life — carbohydrates, proteins, lipids, nucleic acids, vitamins and minerals, and water — each have distinct molecular structures, specific biological functions and identifiable Kenyan food sources. The absence of even one chemical causes a specific, predictable and often visible health consequence. The child eating ugali develops kwashiorkor because maize starch provides carbohydrates but no essential amino acids, so the body cannot synthesise albumin, keratin or

	muscle proteins. The githeri child remains healthy because beans supply essential amino acids that complement the maize carbohydrate.
How does this explain the phenomenon?	Learners explained that food is not interchangeable by quantity alone — its chemical composition determines its biological effect. A diet sufficient in carbohydrates can still cause severe disease if proteins are absent, because each chemical of life performs functions that no other chemical can substitute. This understanding explains the difference between the two Kisumu children and connects molecular biochemistry to community health outcomes across Kenya.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.